# Terminal Pinning in Prediction Markets: Option Pricing Near Event Resolution via Information-Driven Bridge Dynamics


**Abstract**

Prediction-market event futures possess a defining characteristic: Prices converge to binary outcomes $\{0, 1\}$ at a known terminal time $T$. Existing models treat this "pinning", as we refer to it, as an imposed boundary rather than a consequence of the natural dynamics. In this paper, I develop an information-based framework in which the pinning emerges endogenously: Conditional on the unobserved outcome $X_T \in \{0, 1\}$, the price follows a Brownian-bridge process; unconditionally, it is a martingale mixture of two such bridges weighted by the posterior probability of the event. I formulate sharp leading-order asymptotic expansions for implied volatility as the time to maturity tends to zero near event resolution, establish that market completeness requires a traded option despite the single Brownian-motion structure, and verify the three model implied predictions empirically on resolved Polymarket contracts totalling \$6.4B in volume from 2024–2026.



**Keywords:** prediction markets, event derivatives, Brownian bridge, terminal pinning, information-based asset pricing, implied volatility asymptotics, market completeness.

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*Code and data availability.* The Python data-collection script and analysis pipeline used in Section 6 are released as supplementary material accompanying this manuscript and will be archived at a permanent DOI upon acceptance.

# 1 Introduction

A prediction-market event future is a contract that pays \$1 at a known terminal time $T$ if a specific event $E$ occurs, and zero otherwise. The traded price $P_t$ at any earlier time $t \in [0, T]$ takes values in $[0, 1]$ and results in the interpretation:

$$P_t = \mathbb{Q}(E \mid \mathcal{F}_t), \tag{1}$$

that is, the market's current estimate of the probability that the event will occur, computed under an equivalent martingale measure $\mathbb{Q}$ on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{Q})$. Please note that we work throughout with zero interest rates, which simplifies notation without affecting any of the substantive results.

The asset class has expanded rapidly over the past several years. CFTC-regulated venues and decentralized platforms, such as Kalshi and Polymarket respectively, now process substantial trading volumes on major economic, political, and policy events. The launch of perpetual futures by both platforms in April 2026, together with active discussion of listed options, makes the question of how to theoretically and empirically price derivatives written on prediction-market event futures practically consequential.

In the recent literature, two main classes of models have emerged to describe the dynamics of $P_t$. The first is the martingale Jacobi diffusion, also known in population genetics as the Wright–Fisher diffusion, satisfying

$$\mathrm{d}P_t = \sigma\sqrt{P_t(1 - P_t)}\,\mathrm{d}W_t^{\mathbb{Q}}, \tag{2}$$

which respects the bounded state space $[0, 1]$ because its diffusion coefficient vanishes at the two boundaries. The second is the logit jump-diffusion of Dalen [2025], in which the log-odds transform $X_t = \log(P_t/(1 - P_t))$ follows a Lévy-driven stochastic differential equation. This means that the boundedness of $P_t$ then follows from the inverse logit map. Both specifications admit tractable transition densities, produce semi-closed-form prices for European-style payoffs, and can be calibrated to observed Polymarket and Kalshi price histories with acceptable goodness-of-fit at typical horizons.

## 1.1 The terminal pinning problem

Despite their theoretical convenience, both classes of models share a fundamental problem. At resolution time $T$, the price $P_T$ must almost surely be equal to the indicator $X_T \coloneqq \mathbf{1}_{\{E\}} \in \{0, 1\}$, since the contract settles to the result obtained. However, neither model produces this terminal collapse as an endogenous consequence of their dynamics. Under Jacobi diffusion, the price process can reach the boundaries $\{0\}$ and $\{1\}$ in finite time when its parameters lie on the Feller boundary, but the hitting time is random and does not have a necessary relation to the contractual resolution time $T$. Under logit jump-diffusion, the price never reaches $\{0\}$ or $\{1\}$ at all, so the terminal pinning $P_T = X_T$ must be imposed as an exogenous boundary condition without the support of the underlying SDE.

This deficiency manifests itself in three distinct ways. *Mathematically*, a pricing model whose dynamics does not produce the contractually mandated terminal value is internally inconsistent. The standard remedy in the literature is to restrict attention to options maturing at $\tau < T$— to analyze events sufficiently far from event resolution to allow treatment as a continuous-state diffusion problem. This restriction excludes the contracts that have the greatest practical relevance. *Empirically*, the existing diffusion and jump-diffusion models do not generate the elevated and skewed implied-volatility surfaces that one would expect on theoretical grounds in a market converging on a binary outcome without imposing a time-dependent volatility function chosen to fit the data. *Economically*, a substantial fraction of trading activity in event-contingent instruments tends to concentrate in the final phase of a contract's life, when uncertainty about the realized outcome is most acute. The terminal-pinning regime is not a peripheral case to be excluded from analysis; it is the regime in which derivative pricing has the greatest practical bearing.

## 1.2 Contributions

This paper develops a model in which terminal pinning arises endogenously from the price dynamics. The construction draws on the information-based asset-pricing framework of Brody et al. [2008a,b]: rather than specifying the price process directly, we specify an information process $\xi_t$ whose innovations drive the Bayesian updating of the market's posterior belief about the terminal outcome. The price $P_t$ is then defined as that posterior probability. Conditional on the realized outcome $X_T$, the resulting price process is a deterministic transformation of an information-driven Brownian bridge from $P_0$ to $X_T$, with bridge tension scaling with the residual information yet to arrive. Unconditional on $X_T$, the price is a martingale mixture of two such bridges, with weights given by the running posterior probability—which is, by construction, $P_t$ itself.

The paper makes two principal contributions, supported by additional structural and empirical results.

The first principal contribution is an information-driven bridge model for prediction-market dynamics (Section 2). We construct the unconditional law of $P_t$ as a martingale process that pins to $\{0, 1\}$ at time $T$ almost surely. The associated stochastic differential equation has the form

$$\mathrm{d}P_t = \sigma_t(P_t)\,\mathrm{d}W_t^{\mathbb{Q}}, \qquad \sigma_t(p) = \frac{\beta(t)}{\sqrt{\gamma(t)}}\,p(1-p),$$

where $\beta$ and $\gamma$ are deterministic functions controlling the rate and noise level of information arrival. The model is the binary-payoff analogue of the Brownian-bridge information-process framework of Brody, Hughston, and Macrina, which has previously been developed for continuous-payoff cash flows[Brody et al., 2008a,b]; the binary case has, as far as I am aware, not been treated. The construction produces a martingale on $[0, 1]$ with diffusion coefficient $p(1-p)$—quadratic, rather than the square-root scaling of the Jacobi diffusion—which is a structural consequence of the binary-posterior interpretation rather than a modeling choice.

The second principal contribution is a sharp asymptotic expansion of the Black–Scholes implied volatility for short-dated European probability options near event resolution (Section 4.3). For strikes $K \in (0, 1)$, in the joint limit $t, \tau \uparrow T$ with $t < \tau$, $\tau - t \to 0^+$, and $(T-t)/(T-\tau) \to \infty$,

$$\sigma_{\mathrm{imp}}(K, \tau, t, p) \;=\; \frac{\Sigma^*(p, K)}{\sqrt{\tau - t}} \cdot \big(1 + o(1)\big), \tag{3}$$

where $\Sigma^*(p, K)$ is determined explicitly by the unique solution of a Black–Scholes inversion at the limiting bridge price $p(1-K)$ for $K \neq p$, or $p(1-p)$ for $K = p$. This is Theorem 4.2.

In support of these two principal contributions, the paper also establishes the following.

(i) A weak-convergence result with explicit rate, characterizing the rate at which $\mathrm{Law}(P_t)$ collapses to the two-atom distribution $p_0\delta_1 + (1-p_0)\delta_0$ as $t \to T^-$ (Theorem 3.2). The rate is power-law in the residual time $T - t$.

(ii) A finite-sample bound (Lemma 4.5) that connects the asymptotic implied-volatility result to a Black–Scholes inversion of *realized* forward call payoffs, which is what is observable in the absence of a traded prediction-market option panel. The bound decomposes the gap between the realized-payoff inversion and $\Sigma^*$ into a model term that vanishes in the joint limit (28) and a Jensen term of order $1/N$ that is empirically negligible at panel sample sizes.

(iii) A completeness result showing that the market consisting of the event future and money market alone is incomplete in the bridge setting, despite being driven by a single Brownian motion, because the diffusion coefficient $\sigma_t(P_t)$ vanishes in the terminal regime; completeness on $[0, \tau]$ is restored by the addition of a single traded European option whose price process forms a non-degenerate spanning system with the event future (Theorem 5.2).

(iv) An empirical evaluation of the model on a panel of 50 resolved binary Polymarket markets (\$6.4 billion of cumulative trading volume, January 2024 to April 2026, spanning 19 topical categories from US presidential politics to crypto to Federal Reserve policy decisions, Section 6). The data are consistent with all three of the model's central predictions: endogenous pinning at a power-law rate, the limiting payoff $p(1-K)$, and the closed-form constant $\Sigma^*(p, K)$ for short-dated implied volatility. The model's quantitative finite-$\mathcal{L}$ prediction matches the observed level and time-to-resolution dependence of $\hat{\Sigma}_{\text{realized}}/\Sigma^*$ at panel-mean parameters, not just its asymptotic direction. The full data-collection script and analysis pipeline are released alongside the paper.

## 1.3 Relation to existing literature

The information-based asset-pricing framework of Brody et al. [2008a,b, 2011] treats asset prices as conditional expectations of cash flows under a filtration generated by a Brownian-bridge information process. The original setting concerns continuous-payoff cash flows, typically Gaussian or expressible as a deterministic function of a Gaussian state variable. The prediction-market setting we develop here is binary, with $X_T \in \{0, 1\}$, and produces qualitatively different dynamics—most notably the two-atom terminal collapse studied in Section 3. The binary-payoff case appears not to have been developed in the information-based literature, although Rutkowski and Yu [2007] considers a related construction in the context of credit-risk modelling.

The two baseline models against which we compare are the martingale Jacobi diffusion, an instance of the polynomial-diffusion class developed by Filipović and Larsson [2016] and applied to option pricing by Ackerer et al. [2018] in the stochastic-volatility setting, and the logit jump-diffusion of Dalen [2025]. Terminal-pinning structures have appeared in adjacent contexts: in defaultable-bond pricing the bond price collapses to either par or a recovery value at the credit event [Jeanblanc et al., 2009], and in barrier-option pricing under bridge specifications the terminal value is constrained by a barrier-crossing condition [Baldi et al., 1999]. Within the BHM tradition, Rutkowski and Yu [2007] extends the information-based approach to defaultable term structure models in which the recovery rate plays a role analogous to a binary-payoff component; their setting is closer to ours than the original BHM continuous-payoff construction. Our contribution relative to Rutkowski and Yu [2007] is the explicit unconditional-density representation as a martingale mixture of two conditional bridges (Proposition 2.6), the sharp implied-volatility asymptotic with closed-form constant $\Sigma^*(p, K)$ (Theorem 4.2), and the completeness analysis with vanishing diffusion coefficient (Theorem 5.2); none of these appears in the credit-risk literature.

## 1.4 Outline

The remainder of the paper is organized as follows. Section 2 introduces the probability space, the binary event structure, and the information process, and derives both the conditional and unconditional dynamics of $P_t$. Section 3 studies the terminal asymptotics of the unconditional law and proves the principal weak-convergence result. Section 4 develops European option pricing under the bridge dynamics and proves the implied-volatility asymptotics. Section 5 treats hedging and market completeness. Section 6 provides an empirical evaluation against a panel of fifty resolved Polymarket markets. Section 7 concludes. Proofs of the main results, together with reproducible numerical methodology, are collected in Appendices A and B.

## 2 The Information-Bridge Model

### 2.1 Probability space and event structure

Fix a finite horizon $T > 0$ and let $(\Omega, \mathcal{F}, \mathbb{Q})$ be a complete probability space supporting a standard Brownian motion $(B_t)_{t\in[0,T]}$ and a $\{0,1\}$-valued random variable $X_T$ with $\mathbb{Q}(X_T = 1) = p_0 \in (0,1)$, independent of $B$. We interpret $X_T$ as the realized outcome of the binary event $E$, revealed publicly at time $T$. The underlying probability measure $\mathbb{Q}$ is taken throughout to be the risk-neutral measure; under our zero-interest-rate convention, the absence of arbitrage requires the price process $P_t$ to be a $\mathbb{Q}$-martingale with terminal value $X_T$.

### 2.2 The information process

Following the architecture of Brody et al. [2008a], we model the process of information arrival rather than the price process directly. Let $\beta\colon [0,T] \to [0,\infty)$ be a measurable function representing the instantaneous *information rate*, with $\int_0^T \beta(s)\mathrm{d}s = +\infty$ (i.e., a divergent total information accrual; this divergence is what produces terminal pinning). Define the information process

$$\xi_t := \int_0^t \beta(s)X_T\,\mathrm{d}s + \int_0^t \sqrt{\gamma(s)}\,\mathrm{d}B_s, \qquad t \in [0,T], \tag{4}$$

where $\gamma\colon [0,T] \to (0,\infty)$ controls the noise level of incoming signals. The process $\xi_t$ is observed by market participants; the outcome $X_T$ is not observed until $t = T$. Heuristically, $\xi_t$ is a noisy cumulative signal about $X_T$, and the market's belief at time $t$ is the posterior $\mathbb{Q}(X_T = 1 \mid \mathcal{F}_t^\xi)$ where $\mathcal{F}_t^\xi = \sigma(\xi_s : s \le t)$.

*Remark* 2.1. The choice of $\beta$ and $\gamma$ controls the speed of information arrival. A natural baseline is $\beta(s) \equiv \beta_0$ and $\gamma(s) \equiv 1$, in which case $\xi_t$ is a Brownian motion with drift $\beta_0 X_T$. We work with general $\beta, \gamma$ until Section 3, where specific choices are required for the asymptotic analysis.

### 2.3 The price process as a posterior probability

Define the price process by Bayesian updating:

$$P_t := \mathbb{Q}(X_T = 1 \mid \mathcal{F}_t^\xi). \tag{5}$$

**Lemma 2.2.** *Under the dynamics* (4)*, the posterior probability* (5) *admits the representation*

$$P_t = \frac{p_0\exp(Z_t)}{p_0\exp(Z_t) + (1-p_0)}, \qquad Z_t := \int_0^t \frac{\beta(s)}{\gamma(s)}\mathrm{d}\xi_s - \tfrac{1}{2}\int_0^t \frac{\beta(s)^2}{\gamma(s)}\mathrm{d}s. \tag{6}$$

*Proof.* The likelihood ratio of the path $(\xi_s)_{s\le t}$ under $\{X_T = 1\}$ versus $\{X_T = 0\}$ is, by Girsanov's theorem applied to the diffusion observation (4),

$$L_t = \exp\left(\int_0^t \frac{\beta(s)}{\gamma(s)}\mathrm{d}\xi_s - \tfrac{1}{2}\int_0^t \frac{\beta(s)^2}{\gamma(s)}\mathrm{d}s\right) = \exp(Z_t),$$

and Bayes' rule gives $P_t = p_0L_t/(p_0L_t+1-p_0)$, which rearranges to (6); see Liptser and Shiryaev [2001, Theorem 7.12] for the underlying filtering theorem. □

**Theorem 2.3.** *The price process $P_t$ defined in* (5) *is an $\mathcal{F}^\xi$-martingale and satisfies the SDE*

$$\mathrm{d}P_t = \frac{\beta(t)}{\sqrt{\gamma(t)}}P_t(1-P_t)\,\mathrm{d}W_t^{\mathbb{Q}}, \qquad P_0 = p_0, \tag{7}$$

*where*

$$W_t^{\mathbb{Q}} := \int_0^t \frac{1}{\sqrt{\gamma(s)}}\,(\mathrm{d}\xi_s - \beta(s)P_s\,\mathrm{d}s) \tag{8}$$

*is the innovation Brownian motion of the filter under $\mathbb{Q}$.*

*Proof.* See Appendix A.1. $\square$

*Remark* 2.4. The diffusion coefficient in (7) is $\sigma_t(p) = \beta(t)\gamma(t)^{-1/2}p(1-p)$, which scales as $p(1-p)$ rather than the Jacobi $\sqrt{p(1-p)}$. This is the principal qualitative difference: the bridge model has *quadratic* scaling at the boundaries, whereas the Jacobi model has *root* scaling. The bridge model thus has *smaller* instantaneous volatility near the boundaries (because $p(1-p) \le \sqrt{p(1-p)}$ on $[0,1]$), which corresponds to slower diffusive movement once the market is confident—but this is more than compensated by the divergence of $\beta(t)/\sqrt{\gamma(t)}$ as $t \to T$, which produces the pinning. We make this trade-off precise in Section 3.

### 2.4 The conditional bridge representation

The key structural insight is that conditional on the terminal outcome $X_T$, the logit-transformed price has Gaussian increments with a deterministic, divergent drift.

**Proposition 2.5.** *Conditional on $X_T = 1$, the logit-transformed price $L_t := \log(P_t/(1-P_t))$ satisfies*

$$L_t = \log\frac{p_0}{1-p_0} + \int_0^t \frac{\beta(s)^2}{\gamma(s)}\mathrm{d}s + \int_0^t \frac{\beta(s)}{\sqrt{\gamma(s)}}\mathrm{d}\tilde{B}_s, \tag{9}$$

*where $\tilde{B}$ is a $\mathbb{Q}(\cdot \mid X_T = 1)$-Brownian motion. Conditional on $X_T = 0$,*

$$L_t = \log\frac{p_0}{1-p_0} - \int_0^t \frac{\beta(s)^2}{\gamma(s)}\mathrm{d}s + \int_0^t \frac{\beta(s)}{\sqrt{\gamma(s)}}\mathrm{d}\tilde{B}'_s, \tag{10}$$

*with $\tilde{B}'$ a $\mathbb{Q}(\cdot \mid X_T = 0)$-Brownian motion.*

*Proof.* See Appendix A.2. $\square$

### 2.5 The unconditional law

Define the conditional density

$$f_t^{(x)}(p) := \text{density of } P_t \text{ under } \mathbb{Q}(\cdot \mid X_T = x), \qquad x \in \{0,1\},\ p \in (0,1). \tag{11}$$

Under the conditional bridge representation, $f_t^{(x)}$ is the law of a logit-Gaussian random variable with mean and variance given by Proposition 2.5.

**Proposition 2.6.** *The unconditional density of $P_t$ under $\mathbb{Q}$ is*

$$f_t(p) = p_0 f_t^{(1)}(p) + (1-p_0) f_t^{(0)}(p), \qquad p \in (0,1). \tag{12}$$

*Proof.* This is the law of total probability conditional on $X_T \in \{0,1\}$. $\square$

## 3 Terminal Asymptotics

We now analyze the behavior of the unconditional law $f_t$ as $t \to T^-$. From here on we adopt the calibration $\gamma \equiv 1$ and

$$\beta(t) = \frac{\beta_0}{\sqrt{T-t}}, \qquad t \in [0,T), \tag{13}$$

for some $\beta_0 > 0$. This choice makes the integral $\int_0^t \beta(s)^2 \mathrm{d}s = \beta_0^2 \log(T/(T-t))$ divergent at $T$ as required, while keeping the analysis tractable.

### 3.1 The conditional logit process: explicit Gaussian law

Under (13), equations (9)–(10) simplify to give an explicit Gaussian law for $L_t$. Let $\ell_0 := \log(p_0/(1-p_0))$ and write

$$\Lambda_t := \log \frac{T}{T-t}.$$

**Lemma 3.1.** *Under* (13)*, conditional on $X_T = x \in \{0,1\}$, $L_t$ is Gaussian with*

$$L_t \sim \mathcal{N}\big(\mu_t^{(x)}, \sigma_t^2\big), \qquad \mu_t^{(1)} = \ell_0 + \beta_0^2\Lambda_t, \quad \mu_t^{(0)} = \ell_0 - \beta_0^2\Lambda_t, \quad \sigma_t^2 = \beta_0^2\Lambda_t. \tag{14}$$

*Proof.* Substituting $\beta(s) = \beta_0/\sqrt{T-s}$, $\gamma \equiv 1$ into Proposition 2.5, the deterministic drift integral evaluates to $\beta_0^2\Lambda_t$ and the stochastic integral

$$\int_0^t \beta(s)\,\mathrm{d}\tilde{B}_s = \int_0^t \frac{\beta_0}{\sqrt{T-s}}\,\mathrm{d}\tilde{B}_s$$

is a Wiener integral against a deterministic kernel, hence centered Gaussian with variance $\int_0^t \beta(s)^2\mathrm{d}s = \beta_0^2\Lambda_t$. The case $X_T = 0$ is symmetric. □

The signal-to-noise ratio of $L_t$ under $\mathbb{Q}^{(1)} := \mathbb{Q}(\cdot \mid X_T = 1)$ is

$$\frac{|\mu_t^{(1)} - \ell_0|}{\sigma_t} = \frac{\beta_0^2\Lambda_t}{\beta_0\sqrt{\Lambda_t}} = \beta_0\sqrt{\Lambda_t} \;\to\; \infty \quad (t \to T^-),$$

which is the precise quantitative form of pinning.

### 3.2 The principal weak-convergence result

**Theorem 3.2** (Terminal law)**.** *Under the calibration* (13)*, the unconditional law of $P_t$ converges weakly to a two-atom distribution as $t \to T^-$:*

$$P_t \overset{\mathbb{Q}}{\Longrightarrow} p_0\,\delta_1 + (1-p_0)\,\delta_0, \qquad t \to T^-. \tag{15}$$

*The rate of convergence in the bounded-Lipschitz metric is*

$$d_{BL}\big(\mathrm{Law}(P_t),\, p_0\delta_1 + (1-p_0)\delta_0\big) \;\le\; 2\left(\frac{T-t}{T}\right)^{\beta_0^2/2}, \qquad t \in [0,T). \tag{16}$$

*Proof.* See Appendix A.3. □

*Remark* 3.3. The bound (16) is power-law in the residual time $T-t$. The exponent $\beta_0^2/2$ is itself directly testable on data: rearranging (16) via the moment identity in the proof gives $\mathbb{E}[P_t(1-P_t)] \le ((T-t)/T)^{\beta_0^2/2}$, so log-linear regression of the realized $P_t(1-P_t)$ against $\log((T-t)/T)$ identifies $\beta_0$. We exploit this in Section 6.

### 3.3 Convergence under general information-rate calibrations

The closed-form rate (16) relies on the specific calibration (13), $\beta(t) = \beta_0/\sqrt{T-t}$, which makes the conditional logit $L_t$ exactly Gaussian under the time change $\Lambda_t$. We record here the more general result that weak convergence to the two-atom limit holds under any information-rate function $\beta$ satisfying $\int_0^T \beta(s)^2\mathrm{d}s = +\infty$, even when the conditional law of $L_t$ is no longer of closed form.

**Theorem 3.4** (General convergence)**.** *Let $\beta : [0,T] \to [0,\infty)$ be measurable with $\int_0^t \beta(s)^2\mathrm{d}s < \infty$ for all $t < T$ and $\int_0^T \beta(s)^2\mathrm{d}s = +\infty$. Then under the corresponding bridge construction:*

*(i)* $\mathbb{E}[P_t(1-P_t)] \to 0$ *as* $t \to T^-$.

*(ii)* $P_t \overset{\mathbb{Q}}{\Longrightarrow} p_0\delta_1 + (1-p_0)\delta_0$, *with the bounded-Lipschitz rate*

$$d_{BL}\big(\mathrm{Law}(P_t), p_0\delta_1 + (1-p_0)\delta_0\big) \le 2\,\mathbb{E}[P_t(1-P_t)]. \tag{17}$$

*(iii) Conditional on* $X_T = x \in \{0,1\}$, $L_t \to (2x-1)\infty$ *in* $\mathbb{Q}^{(x)}$*-probability.*

*Sketch.* Assume the calibration $\gamma \equiv 1$ adopted in Section 3. By the same construction as in Section 2, $L_t = \log(P_t/(1-P_t))$ is a continuous semimartingale whose conditional law given $X_T = x$ has mean $\ell_0 + (2x-1)\Gamma_t$ and quadratic variation $\Gamma_t$, where $\Gamma_t := \int_0^t \beta(s)^2\,\mathrm{d}s$. The hypothesis $\int_0^T \beta^2\,\mathrm{d}s = +\infty$ is exactly $\Gamma_{T^-} = +\infty$. Under $\mathbb{Q}^{(1)}$, $L_t/\sqrt{\Gamma_t}$ has unit variance with mean $\sqrt{\Gamma_t} \to \infty$, so $L_t \to +\infty$ in $\mathbb{Q}^{(1)}$-probability by Chebyshev. Hence $P_t \to 1$ in $\mathbb{Q}^{(1)}$-probability, and symmetrically $P_t \to 0$ in $\mathbb{Q}^{(0)}$-probability. The moment estimate $\mathbb{E}[P_t(1-P_t)] \to 0$ follows from bounded convergence ($P_t(1-P_t) \le 1/4$) combined with the conditional convergence in probability to zero. Inequality (17) is identical to (33) in the proof of Theorem 3.2. □

The cost of dropping calibration (13) is that the rate (17) is no longer closed-form: $\mathbb{E}[P_t(1-P_t)]$ depends on $\beta(\cdot)$ through the function $\Gamma_t$, and is power-law in $T-t$ only when $\Gamma_t$ has logarithmic growth in $T-t$ — which is the calibration (13) up to a constant. For $\beta(t) = \beta_0/(T-t)^\alpha$ with $\alpha \in (1/2,1)$, $\Gamma_t$ has polynomial growth in $1/(T-t)$, and the resulting rate is exponential in $1/(T-t)^{2\alpha-1}$ rather than power-law. This sensitivity to the functional form of $\beta$ matters empirically; we return to it in Section 6.

### 3.4 The residual continuous component

**Proposition 3.5.** *For any* $\epsilon \in (0,1/2)$,

$$\mathbb{Q}(P_t \in (\epsilon, 1-\epsilon)) \;\le\; \frac{1}{\epsilon(1-\epsilon)}\left(\frac{T-t}{T}\right)^{\beta_0^2/2}. \tag{18}$$

*Proof.* By Markov's inequality applied to $P_t(1-P_t) \ge \epsilon(1-\epsilon)\mathbf{1}_{\{P_t\in(\epsilon,1-\epsilon)\}}$,

$$\mathbb{Q}(P_t \in (\epsilon, 1-\epsilon)) \le \frac{\mathbb{E}[P_t(1-P_t)]}{\epsilon(1-\epsilon)},$$

and the bound (18) follows from the estimate $\mathbb{E}[P_t(1-P_t)] \le ((T-t)/T)^{\beta_0^2/2}$ established in the proof of Theorem 3.2. □

## 4 Option Pricing Under the Bridge Model

### 4.1 European call prices

A European probability call with strike $K \in (0,1)$ and maturity $\tau \in (t,T]$ pays $(P_\tau - K)^+$ at $\tau$. Under the zero-interest-rate convention,

$$c(t,p;\tau,K) = \mathbb{E}^{\mathbb{Q}}\big[(P_\tau - K)^+ \,\big|\, P_t = p\big]. \tag{19}$$

**Proposition 4.1.** *Under the bridge model with calibration* (13)*, conditional on* $P_t = p$ *and* $X_T = x$*,* $L_\tau$ *is Gaussian with*

$$L_\tau \mid \{P_t = p, X_T = x\} \;\sim\; \mathcal{N}\big(m^{(x)}(t,\tau,p),\, v(t,\tau)\big), \tag{20}$$

*where, writing* $\mathcal{L} := \log \frac{T-t}{T-\tau}$*,*

$$m^{(1)}(t,\tau,p) = \mathrm{logit}(p) + \beta_0^2\mathcal{L}, \quad m^{(0)}(t,\tau,p) = \mathrm{logit}(p) - \beta_0^2\mathcal{L}, \quad v(t,\tau) = \beta_0^2\mathcal{L}, \tag{21}$$

*and the call price decomposes as*

$$c(t,p;\tau,K) = p\,c^{(1)} + (1-p)\,c^{(0)}, \tag{22}$$

*with*

$$c^{(x)} = \mathbb{E}\Big[\big(\Phi(m^{(x)} + \sqrt{v}\,Z) - K\big)^+\Big], \qquad Z \sim \mathcal{N}(0,1), \tag{23}$$

*and* $\Phi(\ell) := e^\ell/(1+e^\ell)$ *the logistic function.*

*Proof.* Conditional on $P_t = p$, by the strong Markov property of $L$ derived in Proposition 2.5, the increment $L_\tau - L_t$ is independent of $\mathcal{F}_t^\xi$. Substituting $\beta(s) = \beta_0/\sqrt{T-s}$, the integrals over $[t,\tau]$ evaluate to $\beta_0^2\mathcal{L}$ for both the deterministic drift and the variance of the Wiener integral, giving (21). The decomposition (22) is the law of total expectation. □

### 4.2 Limit of the call price as $\tau \to T^-$

A central observation drives the implied-volatility analysis. Because $P$ is a $\mathbb{Q}$-martingale converging almost surely to $X_T \in \{0,1\}$, the call payoff converges almost surely:

$$(P_\tau - K)^+ \longrightarrow (X_T - K)^+ = (1-K)\mathbf{1}_{\{X_T=1\}}, \qquad \tau \to T^-,$$

for $K \in (0,1)$. Conditioning on $P_t = p$ and using $\mathbb{E}[\mathbf{1}_{\{X_T=1\}} \mid P_t = p] = p$, dominated convergence yields

$$\lim_{\tau\to T^-} c(t,p;\tau,K) \;=\; p(1-K), \qquad K \in (0,1). \tag{24}$$

The limiting price $p(1-K)$ is strictly interior to the Black–Scholes no-arbitrage interval $(\max(p-K,0),\, p)$ for any $K \in (0,1)$. This boundedness of the limiting price is what determines the implied-volatility asymptotic.

### 4.3 Implied volatility asymptotics

The Black–Scholes implied volatility is defined as the value $\sigma_{\mathrm{imp}}(K,\tau,t,p)$ that solves

$$c_{BS}(p,K,\sigma_{\mathrm{imp}},\tau-t) = c(t,p;\tau,K), \tag{25}$$

where $c_{BS}(p,K,\sigma,\Delta) = p\Phi_{\mathcal{N}}(d_1) - K\Phi_{\mathcal{N}}(d_2)$ is the Black–Scholes call price at zero interest with $d_{1,2} = (\log(p/K) \pm \frac{1}{2}\sigma^2\Delta)/(\sigma\sqrt{\Delta})$ and $\Phi_{\mathcal{N}}$ the standard normal cdf. We adopt this convention because it is the practitioner standard for bounded-underlying derivatives, with the understanding that $\sigma_{\mathrm{imp}}$ is a quoting convention rather than a structural parameter.

It will be convenient to work with the *normalized total implied variance*

$$\Sigma := \sigma_{\mathrm{imp}}\sqrt{\tau - t},$$

and to view $c_{BS}$ as a function of $\Sigma$ alone (since the dependence on $\sigma$ and $\Delta$ enters only through this product):

$$\widetilde{c}_{BS}(p,K,\Sigma) := p\Phi_{\mathcal{N}}(d_1^\Sigma) - K\Phi_{\mathcal{N}}(d_2^\Sigma), \qquad d_{1,2}^\Sigma = \frac{\log(p/K) \pm \Sigma^2/2}{\Sigma}.$$

For each fixed $p, K \in (0,1)$, the map $\Sigma \mapsto \widetilde{c}_{BS}(p,K,\Sigma)$ is strictly increasing on $(0,\infty)$ from $\max(p-K,0)$ to $p$.

**Theorem 4.2** (Implied volatility asymptotics)**.** *Fix $p \in (0,1)$ and $K \in (0,1)$. For $K \neq p$, define $\Sigma^*(p,K)$ as the unique solution in $(0,\infty)$ of*

$$\widetilde{c}_{BS}(p,K,\Sigma^*(p,K)) = p(1-K). \tag{26}$$

*For $K = p$, define*

$$\Sigma^*(p,p) = 2\Phi_{\mathcal{N}}^{-1}\left(1 - \tfrac{p}{2}\right). \tag{27}$$

*Consider any sequence $(t_n, \tau_n)$ with $t_n < \tau_n < T$ such that, as $n \to \infty$,*

$$\tau_n - t_n \to 0^+ \qquad \textit{and} \qquad \mathcal{L}_n := \log \frac{T - t_n}{T - \tau_n} \to \infty. \tag{28}$$

*Then*

$$\sigma_{\mathrm{imp}}(K, \tau_n, t_n, p) \;=\; \frac{\Sigma^*(p,K)}{\sqrt{\tau_n - t_n}} \big(1 + o(1)\big). \tag{29}$$

*The convergence is uniform over any subset of pairs $(t,\tau)$ on which $\mathcal{L} = \log((T-t)/(T-\tau))$ is bounded below by a divergent function of $n$.*

*Proof.* See Appendix A.4. □

*Remark* 4.3. Theorem 4.2 states that $\sigma_{\mathrm{imp}}\sqrt{\tau - t} \to \Sigma^*(p,K)$ as the option becomes both short-dated and close to event resolution, in the precise sense of (28). The two conditions $\tau_n - t_n \to 0$ and $\mathcal{L}_n \to \infty$ are independent: $\tau_n - t_n \to 0$ alone (e.g., shrinking the option's tenor at a fixed valuation date $t < T$) does not produce the asymptotic; one also requires the valuation date $t_n$ to approach $T$ fast enough that the ratio $(T - t_n)/(T - \tau_n)$ grows without bound. This is the natural near-resolution short-dated regime relevant for empirical work, and is what (28) formalizes. The constant $\Sigma^*(p,K)$ is the Black–Scholes total implied variance that prices the limiting bridge call $p(1-K)$ exactly. Several further observations are worth recording:

(i) Because $p(1-K) \in (0,p)$ for $K \in (0,1)$, the equation (26) has a unique finite solution, and consequently $\sigma_{\mathrm{imp}} \to \infty$ in the joint limit: the divergence is the standard short-dated $1/\sqrt{\tau - t}$ scaling, with a model-specific multiplicative constant $\Sigma^*(p,K)$.

(ii) Equation (26) contains *no* free parameters of the bridge model: $\Sigma^*(p,K)$ depends only on the structurally determined limiting payoff $p(1-K)$. The information rate $\beta_0$ controls how quickly the limit is approached, but not the limit itself. This makes Theorem 4.2 a particularly sharp model prediction.

(iii) No additional divergent factor in $\log(T/(T-\tau))$ appears, contrary to what a naive analysis based on the conditional logit drift might suggest. The drift divergence is exactly absorbed in the saturation of the call price at $p(1-K)$.

*Remark* 4.4 (ATM constant). The ATM constant $\Sigma^*(p,p)$ admits the closed form (27) because at $K = p$, $\widetilde{c}_{BS}(p,p,\Sigma) = p[2\Phi_{\mathcal{N}}(\Sigma/2) - 1]$, and equation (26) becomes $p[2\Phi_{\mathcal{N}}(\Sigma/2) - 1] = p(1-p)$, i.e. $\Phi_{\mathcal{N}}(\Sigma/2) = 1 - p/2$.

### 4.4 Smile shape

The constant $\Sigma^*(p,K)$ as a function of $K$ encodes the asymptotic smile shape under the bridge model. Standard implicit differentiation of (26) yields

$$\frac{\partial \Sigma^*}{\partial K}(p,K) = -\frac{\partial_K \widetilde{c}_{BS}(p,K,\Sigma^*) - \partial_K[p(1-K)]}{\partial_\Sigma \widetilde{c}_{BS}(p,K,\Sigma^*)} = \frac{p - \Phi_{\mathcal{N}}(d_2^{\Sigma^*})}{\partial_\Sigma \widetilde{c}_{BS}(p,K,\Sigma^*)},$$

which is strictly positive for $\Phi_{\mathcal{N}}(d_2^{\Sigma^*}) < p$, i.e., for OTM strikes. The smile rises steeply at low strikes (high $\Sigma^*$) and falls less rapidly at high strikes, producing the characteristic asymmetric shape illustrated in Section 6.

### 4.5 Comparison with the Jacobi diffusion

Under the martingale Jacobi diffusion (2), the call price at $\tau \to T^-$ does not approach $p(1-K)$; instead the law of $P_T$ has a continuous density on $[0,1]$ (when $\sigma$ does not push parameters to the Feller boundary) and the call price approaches $\mathbb{E}[(P_T - K)^+]$ which is bounded away from $p(1-K)$ in general. The Jacobi-implied volatility is therefore bounded above by a finite constant $\Sigma^*_{\mathrm{Jac}}(p,K)$ determined by the Jacobi terminal law, which differs from $\Sigma^*(p,K)$ of the bridge model.

### 4.6 From option prices to realized payoffs: a finite-sample bound

The empirical evaluation in Section 6 substitutes a Black–Scholes inversion of *realized forward call payoffs* for a Black–Scholes inversion of *traded option prices*, since a panel of traded prediction-market option prices does not yet exist at the cross-sectional and time-series depth required for the asymptotic test. We record here a quantitative bound on the difference between the two inversions that justifies the substitution under explicit conditions, and that we use in Section 6.4 for a quantitative reconciliation between the model and the panel data.

**Lemma 4.5** (Realized-payoff inversion bound)**.** *Fix $p, K \in (0,1)$ with $K \neq p$, and $\Delta > 0$. Let $\hat{c}_N := N^{-1}\sum_{i=1}^N (P_{t_i+\Delta} - K)^+$ be the sample mean of realized forward payoffs from $N$ pairs $(t_i, t_i+\Delta)$ drawn i.i.d. with $P_{t_i} = p$ under $\mathbb{Q}$, and define the realized-payoff total implied variance*

$$\widehat{\Sigma}_N := \widetilde{c}_{BS}^{-1}(p, K, \hat{c}_N).$$

*Let $\mu := \mathbb{E}^{\mathbb{Q}}[(P_{t+\Delta} - K)^+ \mid P_t = p] = c(t, p; t+\Delta, K)$ denote the true conditional mean. Then there exist constants $C_{p,K}, M_{p,K} > 0$ depending only on $p$ and $K$ such that*

$$\left|\mathbb{E}\widehat{\Sigma}_N - \Sigma^*(p,K)\right| \leq \underbrace{\frac{C_{p,K}}{p\,\varphi_{\mathcal{N}}(d_1^{\Sigma^*})}\, e^{-\beta_0^2\mathcal{L}/2}}_{\textit{model term}} + \underbrace{\frac{M_{p,K}}{N}\,(1-K)^2}_{\textit{Jensen term}} + o\big(N^{-1}\big) + o\big(e^{-\beta_0^2\mathcal{L}/2}\big), \tag{30}$$

*where $\mathcal{L} = \log((T-t)/(T-t-\Delta))$. The model term controls the gap between the bridge-model call price and its asymptotic limit $p(1-K)$; the Jensen term controls the bias from inverting Black–Scholes at a sample mean rather than at its expectation.*

*Proof.* By Theorem 2.3, $\hat{c}_N \to \mu$ in $L^2$ at rate $O(N^{-1/2})$. The map $g(\cdot) := \widetilde{c}_{BS}^{-1}(p, K, \cdot)$ is $C^\infty$ on a neighbourhood of $\mu$ for $\mu \in (\max(p-K, 0), p)$, with $g'(\mu) = 1/[p\,\varphi_{\mathcal{N}}(d_1^{\mu})]$ as in (38). Taylor expansion of $g$ at $\mu$ gives

$$g(\hat{c}_N) = g(\mu) + g'(\mu)(\hat{c}_N - \mu) + \tfrac{1}{2}g''(\xi_N)(\hat{c}_N - \mu)^2,$$

for some $\xi_N$ between $\hat{c}_N$ and $\mu$. Taking expectations, using $\mathbb{E}[\hat{c}_N] = \mu$, and bounding $\mathrm{Var}(\hat{c}_N) \leq N^{-1}(1-K)^2$ since payoffs lie in $[0, 1-K]$,

$$\left|\mathbb{E}g(\hat{c}_N) - g(\mu)\right| \leq \tfrac{1}{2}\sup_{c\in B}|g''(c)|\,\frac{(1-K)^2}{N} + o(N^{-1}), \tag{31}$$

where $B$ is a neighbourhood of $\mu$ that contains $\hat{c}_N$ with high probability. Setting $M_{p,K} = \frac{1}{2}\sup_{c\in B}|g''(c)|$ gives the Jensen term. Combining (31) with the model-term bound $|g(\mu) - \Sigma^*(p,K)| \leq C_{p,K}(p\varphi_{\mathcal{N}})^{-1}e^{-\beta_0^2\mathcal{L}/2}$ from the proof of Theorem 4.2 (equation (40), with $\mu$ playing the role of $c(t,p;\tau,K)$ and $\mathcal{L}$ the same role) and applying the triangle inequality yields (30). □

*Remark* 4.6 (Magnitudes). For typical empirical magnitudes from Section 6 ($p \in [0.1, 0.5]$, $K = p$, $\beta_0 \sim 1$, $N \in [50, 500]$ pairs per cell), numerical evaluation of (30) at the parameter values relevant for the panel gives a Jensen term of order $10^{-3}$, whereas the model term at

$T - t = 2.25$ d, $\Delta = 1$ d (so $\mathcal{L} = \log(2.25/1.25) \approx 0.59$) is of order 0.7, and at $T - t = 15$ d, $\Delta = 1$ d (so $\mathcal{L} = \log(15/14) \approx 0.07$) is of order 0.97. The Jensen term is empirically negligible at the panel sample sizes; the model term is the dominant contribution and is itself the testable prediction. Lemma 4.5 thus reduces the realized-payoff test to a test of (37) at finite $\mathcal{L}$, with quantitative error control.

# 5 Hedging and Completeness

## 5.1 Filtration considerations

The price process $P_t$ is adapted to $\mathcal{F}^\xi$, the filtration of the information process. Equality of the price filtration with the information filtration must be argued via the path-functional sufficient statistic $Z_t$ rather than the value $\xi_t$ alone, because $P_t$ depends on the entire path of $\xi$ up to time $t$.

**Proposition 5.1.** *Under the bridge model with calibration* (13)*, $\mathcal{F}^P_t = \mathcal{F}^\xi_t$ for all $t \in [0, T)$.*

*Proof.* By Lemma 2.2, $P_t = p_0 e^{Z_t}/(p_0 e^{Z_t} + (1 - p_0))$, a strictly monotone smooth bijection of $Z_t$. Hence $\mathcal{F}^P_t = \mathcal{F}^Z_t$. Since $Z_t = \int_0^t (\beta(s)/\gamma(s))\mathrm{d}\xi_s - \frac{1}{2}\int_0^t (\beta(s)^2/\gamma(s))\mathrm{d}s$ is $\mathcal{F}^\xi$-measurable, $\mathcal{F}^Z_t \subseteq \mathcal{F}^\xi_t$. Conversely, because $\theta(s) := \beta(s)/\gamma(s)$ is strictly positive and continuous on $[0, t]$ for $t < T$, the Itô integral $\xi \mapsto \int_0^\cdot \theta(s)\mathrm{d}\xi_s$ is invertible pathwise (by the standard inverse-Itô-integral construction with deterministic positive integrand), giving $\mathcal{F}^\xi_t \subseteq \mathcal{F}^Z_t$. Therefore $\mathcal{F}^P_t = \mathcal{F}^Z_t = \mathcal{F}^\xi_t$. ☐

The innovation Brownian motion $W^\mathbb{Q}$ defined in (8) satisfies $\mathcal{F}^{W^\mathbb{Q}}_t = \mathcal{F}^\xi_t$, so traders observing $P_t$ have full access to the relevant noise.

## 5.2 Market completeness

We turn to whether contingent claims can be replicated by trading the event future and a money market account alone. The result is more delicate than naive dimension-counting suggests, because the diffusion coefficient $\sigma_t(p) = \beta(t)p(1 - p)$ becomes small as $P_t$ approaches $\{0, 1\}$, which happens with high probability as $\tau \to T$.

**Theorem 5.2** (Completeness with one option)**.** *Fix $\tau \in (0, T)$ and consider the market consisting of:*

*(i) a money market account $B_t \equiv 1$,*

*(ii) the event future with price $P_t$ following the bridge dynamics* (7)*,*

*(iii) one European probability option, with maturity $\tau$ and $C^2$ payoff $h : (0, 1) \to \mathbb{R}$ satisfying $h''(p) > 0$ on $(0, 1)$ and $\inf_{p\in[\epsilon,1-\epsilon]} |h'(p)| > 0$ for every $\epsilon \in (0, 1/2)$, traded at the no-arbitrage price $U_t := \mathbb{E}^\mathbb{Q}[h(P_\tau) \mid \mathcal{F}^\xi_t]$.*

*The market is complete on $[0, \tau]$: every $\mathcal{F}^\xi_\tau$-measurable bounded contingent claim admits a self-financing replicating strategy in $(B, P, U)$, unique up to indistinguishability.*

*Proof.* See Appendix A.5. ☐

*Remark* 5.3. The reason an additional traded asset is required is that, although the bridge model is driven by a single Brownian motion $W^\mathbb{Q}$, the diffusion coefficient $\sigma_t(P_t)$ approaches zero as $P_t \to \{0, 1\}$. While $\{(t, \omega) : P_t(\omega) \in \{0, 1\}\}$ has Lebesgue measure zero on $[0, \tau] \times \Omega$ for $\tau < T$, the candidate replicating strategy $\theta^P = \phi/\sigma_t(P_t)$ may fail admissibility (square-integrability of wealth) near the boundaries. The traded option's price process $U_t$ provides a non-degenerate spanning vector $(\sigma_t(P_t), \partial_p u(t, P_t)\sigma_t(P_t))$, where $u(t,p) := \mathbb{E}^{\mathbb{Q}}[h(P_\tau) \mid P_t = p]$ solves the Kolmogorov backward equation, allowing the boundary contributions to be absorbed into the option position.

The implication for derivative-market design is direct: any prediction-market exchange supporting hedging in the terminal-pinning regime should list at least one option product alongside the underlying event future.

### 5.3 Pricing in the incomplete-market regime

When the market consists only of $(B, P)$ and is incomplete in the sense of Theorem 5.2, contingent claims that are not perfectly replicable by trading $P$ admit a range of arbitrage-free prices, bracketed by the upper and lower hedging prices. Within this range, the standard literature offers several candidate pricing kernels: the minimal-entropy martingale measure of Frittelli [2000], the minimal martingale measure of Föllmer and Sondermann [1986], and utility-indifference prices in the sense of Henderson and Hobson [2009]. Each requires a reference measure or a utility specification, and each will in general differ from the construction measure $\mathbb{Q}$ used to define the bridge dynamics.

The substantive question is what reference measure is appropriate when an econometric estimate of the historical (real-world) measure $\mathbb{P}$ is available from observed price paths. Under $\mathbb{P}$, the bridge parameter $\beta_0$ would be re-estimated from data; under $\mathbb{Q}$ as constructed, $\beta_0$ is a primitive of the model. The relation between the two is governed by the Girsanov density transforming $\mathbb{P}$ into a martingale measure, and the minimal-entropy choice picks the EMM closest to $\mathbb{P}$ in $D_{KL}$ distance. Computing this projection explicitly under the bridge specification is straightforward in principle but tangential to our main results, and we do not pursue it here.

## 6 Empirical Evaluation on Polymarket Data

We evaluate the bridge model on a set of resolved binary Polymarket markets. The aim is not formal calibration but a falsification test of three model-implied predictions:

(P1) *Endogenous pinning at a power-law rate.* Theorem 3.2 predicts $\mathbb{E}[P_t(1-P_t)] \le ((T-t)/T)^{\beta_0^2/2}$.

(P2) *Limiting payoff $p(1-K)$.* Equation (24) predicts $\mathbb{E}[(P_\tau - K)^+ \mid P_t = p] \to p(1-K)$ as $\tau \to T^-$.

(P3) *Asymptotic implied-variance constant.* Theorem 4.2 predicts that, in the joint limit (28), the Black–Scholes total implied variance $\Sigma = \sigma_{\text{imp}}\sqrt{\tau - t}$ that prices a European call at maturity $\tau$ converges to the closed-form constant $\Sigma^*(p, K)$ defined by (26)–(27).

**Empirical proxy for $\Sigma^*$.** A direct test of (P3) requires a panel of traded option prices, which does not yet exist for prediction markets at the necessary cross-sectional and time-series depth. We therefore construct an empirical proxy by applying Black–Scholes inversion to *realized forward call payoffs*. For each pair $(t, t+\Delta)$ in a market's history, we compute the realized payoff $\hat{c} = (P_{t+\Delta} - K)^+$ and define $\hat{\Sigma}_{\text{realized}}(t, K, \Delta) := \widetilde{c}_{BS}^{-1}(P_t, K, \hat{c})$. Lemma 4.5 bounds the gap between this proxy and $\Sigma^*(p, K)$ by the sum of two contributions: a model term of order $e^{-\beta_0^2 \mathcal{L}/2}$ that vanishes only in the joint limit (28), and a Jensen term of order $1/N$ from the convexity of Black–Scholes inversion. The model term is the dominant contribution at the residual times available in the panel; the Jensen term is empirically negligible (Remark 4.6). The proxy is therefore strictly weaker than a test on traded option prices, but the difference between the two is itself quantified by the lemma, and Section 6.4 below uses this control to convert the empirical comparison into a quantitative reconciliation. Validating Theorem 4.2 on traded prediction-market option prices, once those become available at sufficient depth, is the most natural follow-up empirical exercise.

## 6.1 Data

We use the Polymarket Gamma and CLOB APIs to retrieve the price history of the YES outcome token for a panel of 50 resolved binary markets, drawn from those with at least $500,000 of cumulative volume and at least 20% of ticks falling in the open interval $[0.05, 0.95]$. The panel was constructed by paging through closed markets in volume-descending order and applying topic-based deduplication (at most 6 markets per topic) and time-based deduplication (at most 10 markets per resolution week) to enforce diversity. The resulting panel transacts $6.4 billion of cumulative volume and spans 19 distinct topical categories: US presidential politics, crypto, sports (MLB, soccer, F1), Federal Reserve policy decisions, Iranian and Russian geopolitics, Chinese policy, government shutdowns, technology (AI, Tesla self-driving), entertainment (Oscars), and others.

Polymarket's CLOB tick fidelity for resolved markets depends on age: very recent markets are retained at 1-minute resolution, while older resolved markets are retained at 720-minute (12-hour) resolution. Our data-collection pipeline cascades through fidelities $\{1, 10, 60, 360, 720, 1440\}$ minutes for each market, accepting the finest available resolution that returns at least 50 ticks. Since the bridge model's predictions concern integrals of price over time (quadratic variation, realized payoffs), the heterogeneity of fidelity across the panel is not a model-relevant distortion, but it does motivate the cross-market pooling strategy of Section 6.3.

For each market we define the *effective resolution time* $T_{\text{eff}}$ as the earliest tick at which the YES price first crossed and remained within 0.01 of its terminal value, capped at the contractual end date. The data-collection script and analysis pipeline are released alongside the paper.

Table 1 reports descriptive statistics for the 12 highest-volume markets in the panel; the full panel is available with the supplementary materials. The panel contains 22 markets that resolved Yes and 28 that resolved No; starting probabilities $p_{\text{init}}$ span the range $[0.005, 0.65]$.

**Table 1:** The twelve highest-volume markets in the panel, ordered by cumulative trading volume. The full panel of fifty markets is available in the supplementary materials. $p_{\text{init}}$ is the YES price at the start of the data window. $T_{\text{eff}}$ is the duration from the start of the data window to the effective resolution time. Interior fraction is the proportion of ticks with $P_t \in [0.05, 0.95]$. $\hat{\beta}_0$ is the QV-based estimate of the information-rate parameter (Section 6.2).

| Question | Topic | Vol. ($M) | Outcome | $p_{\text{init}}$ | $T_{\text{eff}}$ (d) | Int. frac. | $\hat{\beta}_0$ |
|---|---|---|---|---|---|---|---|
| Will Donald Trump win the 2024 US Presidential E... | trump | 1531 | Yes | 0.500 | 305 | 1.00 | 0.44 |
| Will Kamala Harris win the 2024 US Presidential ... | harris | 1037 | No | 0.035 | 18 | 0.41 | 0.31 |
| Will Donald Trump be inaugurated? | trump | 400 | Yes | 0.575 | 62 | 0.33 | 1.28 |
| US forces enter Iran by April 30? | iran | 269 | Yes | 0.635 | 7 | 0.68 | 0.81 |
| Will Zelenskyy wear a suit before July? | russia | 242 | No | 0.095 | 38 | 0.59 | 1.83 |
| Fed decreases interest rates by 50+ bps after Ja... | fed | 235 | No | 0.120 | 106 | 0.49 | 1.12 |
| US x Iran ceasefire by April 7? | iran | 174 | Yes | 0.095 | 9 | 0.37 | 1.15 |
| Kamala Harris wins the popular vote? | harris | 164 | No | 0.500 | 23 | 0.42 | 2.01 |
| Fed decreases interest rates by 50+ bps after De... | fed | 162 | No | 0.110 | 123 | 0.63 | 0.93 |
| US government shutdown Saturday? | shutdown | 157 | Yes | 0.285 | 78 | 0.95 | 1.89 |
| Will Zohran Mamdani win the 2025 NYC mayoral ele... | election | 143 | Yes | 0.095 | 195 | 0.95 | 1.12 |
| Khamenei out as Supreme Leader of Iran by Februa... | iran | 131 | Yes | 0.355 | 44 | 0.63 | 0.79 |

Across the full 50-market panel, the QV-based estimator yields $\hat{\beta}_0$ values with mean 1.19, median 1.11, and standard deviation 0.50, with all values in the range $[0.31, 2.14]$. The dispersion across markets is consistent with $\beta_0$ being a market-specific parameter rather than a universal constant; what is striking is that all values are of order unity — not, for instance, of order 0.01

or 100 — across markets that span more than three orders of magnitude in trading volume and the full range of resolution outcomes.

### 6.2 Test of P1 (pinning rate)

For each market we estimate $\beta_0$ by realized quadratic variation of the logit-transformed price. Under the bridge dynamics with calibration (13), the conditional law of $L_t = \log(P_t/(1-P_t))$ given $X_T$ is Gaussian with quadratic variation $\langle L\rangle_{[t_0,t_1]} = \beta_0^2(\Lambda_{t_1} - \Lambda_{t_0})$, where $\Lambda_t = \log(T_{\text{eff}}/(T_{\text{eff}} - t))$ is the cumulative information process. Computing the realized quadratic variation

$$\widehat{QV} = \sum_{i=0}^{N-1}(L_{t_{i+1}} - L_{t_i})^2 \tag{32}$$

over the available pre-pinning window $[t_0, T_{\text{eff}} - \delta]$ (with $\delta = 5$ minutes to avoid the singularity in $\Lambda_t$ at $T_{\text{eff}}$) and dividing by the corresponding window in $\Lambda$ yields $\hat{\beta}_0 = \sqrt{\widehat{QV}/(\Lambda_{T_{\text{eff}}-\delta} - \Lambda_{t_0})}$. Across all 50 markets in the panel, the fitted $\hat{\beta}_0$ values lie in the range $[0.31, 2.14]$, with mean 1.19 and median 1.11 (s.d. 0.50). All values are of order unity, consistent with the bridge mechanism being the right qualitative description of the dynamics; the 50% dispersion across markets is consistent with $\beta_0$ being market-specific (depending on the information-flow rate of the underlying event).

A complementary test of the upper-bound power-law form (16), $\mathbb{E}[P_t(1-P_t)] \leq ((T-t)/T)^{\beta_0^2/2}$, can be performed by pooled regression. Figure 1 shows the pooled scatter of $\log P_t(1-P_t)$ against $\log((T_{\text{eff}} - t)/T_{\text{eff}})$ across all markets in the panel, with the OLS fit yielding a slope of 0.27. Under the bound (16), the pooled slope is an estimate of (an upper bound on) $\beta_0^2/2$, which corresponds to an effective $\beta_0$ of 0.74. This is within the range of QV-based estimates and consistent with the fact that (16) is an inequality: the QV estimator captures the bridge dynamics exactly, while the OLS slope captures the conditional Gaussian moment generating function evaluated at the boundary, which dominates only when $L_t$ is sufficiently far from the mean. The pooled OLS slope is therefore expected to be smaller than $\beta_0^2/2$ at typical $\hat{\beta}_0$ in the QV-based panel.

A complementary visualization sharpens the qualitative picture. Figure 2 plots the empirical logit-transformed prices $L_t$ for the six highest-volume markets in the panel, alongside the bridge-model conditional mean $\mu_t^{(x)} = \ell_0 + (2x-1)\hat{\beta}_0^2\Lambda_t$ and the $\pm 2$-standard-deviation envelope $\mu_t^{(x)} \pm 2\hat{\beta}_0\sqrt{\Lambda_t}$ predicted by Lemma 3.1. The diverging mean curve characteristic of the bridge ($\mu_t^{(x)} \to \pm\infty$ as $t \to T_{\text{eff}}$) is visible across markets that span the 2024 US presidential election (collective volume \$2.6 billion), the early-2026 Iran policy markets, the Russia–Ukraine market, and the Federal Reserve interest-rate markets.

### 6.3 Test of P2 and P3 (limiting payoff and $\Sigma^*$)

For tests P2 and P3 we pool $(t, t+\Delta)$ pairs across all 50 markets in the panel, with $\Delta = 1$ day. Within each market, all such pairs with $t+\Delta < T_{\text{eff}}$ are formed (subject to a fidelity-aware temporal tolerance), the realized payoff is computed at strike offsets $K - p_t \in \{-0.10, -0.05, 0, +0.05, +0.10\}$, and the BS-implied total variance proxy $\hat{\Sigma}_{\text{realized}} = c_{BS}^{-1}(\hat{c}; p_t, K, \Delta)\sqrt{\Delta}$ is recovered as defined in the proxy paragraph above. Each market yields up to $5 \times 6 = 30$ cell observations indexed by (time-to-resolution bin, strike offset). We then pool ratios $\hat{\Sigma}_{\text{realized}}/\Sigma^*(p_t, K)$ across markets within each (time-to-resolution bin, strike offset) cell. The number of pooled markets per cell ranges from 6 to 28 in the cells with sufficient power for inference.

Table 2 reports the cross-market mean and standard error of the ratio $\hat{\Sigma}_{\text{realized}}/\Sigma^*$ in each cell. The bridge model predicts that the ratio approaches 1 as $T_{\text{eff}} - t \to 0$ within the joint-limit regime (28). Detailed methodology is given in Appendix B.

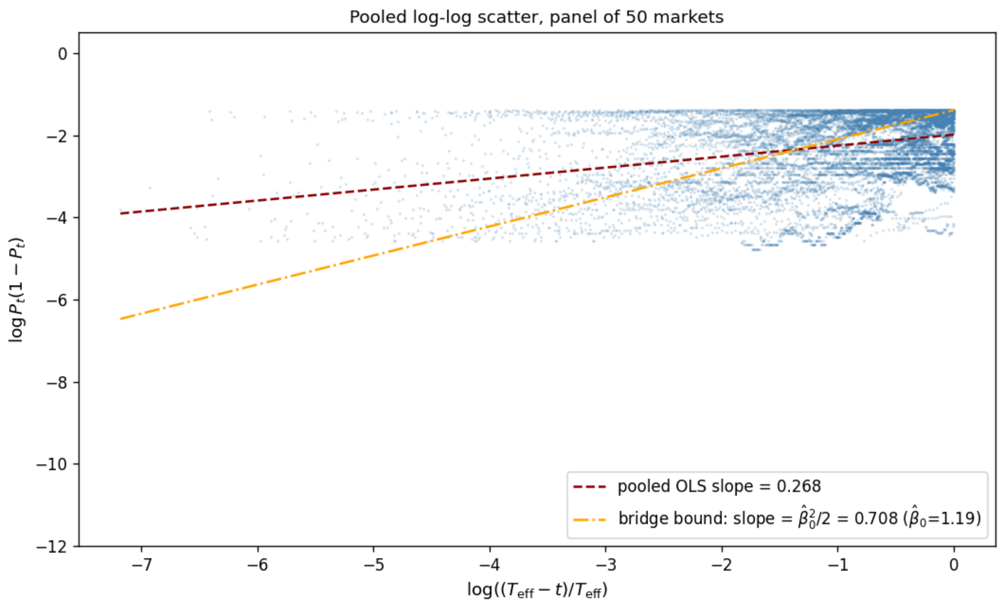


**Figure 1:** Pooled log-log scatter of $P_t(1-P_t)$ against $(T_{\text{eff}}-t)/T_{\text{eff}}$ across the 50-market panel. The pooled OLS slope is 0.27 (dashed red), corresponding to an effective $\beta_0$ of 0.74 under the bound $\mathbb{E}[P_t(1-P_t)] \leq ((T-t)/T)^{\beta_0^2/2}$. The bridge upper bound at the panel-mean $\hat{\beta}_0 = 1.19$ has slope $\hat{\beta}_0^2/2 = 0.71$ (dot-dashed orange).

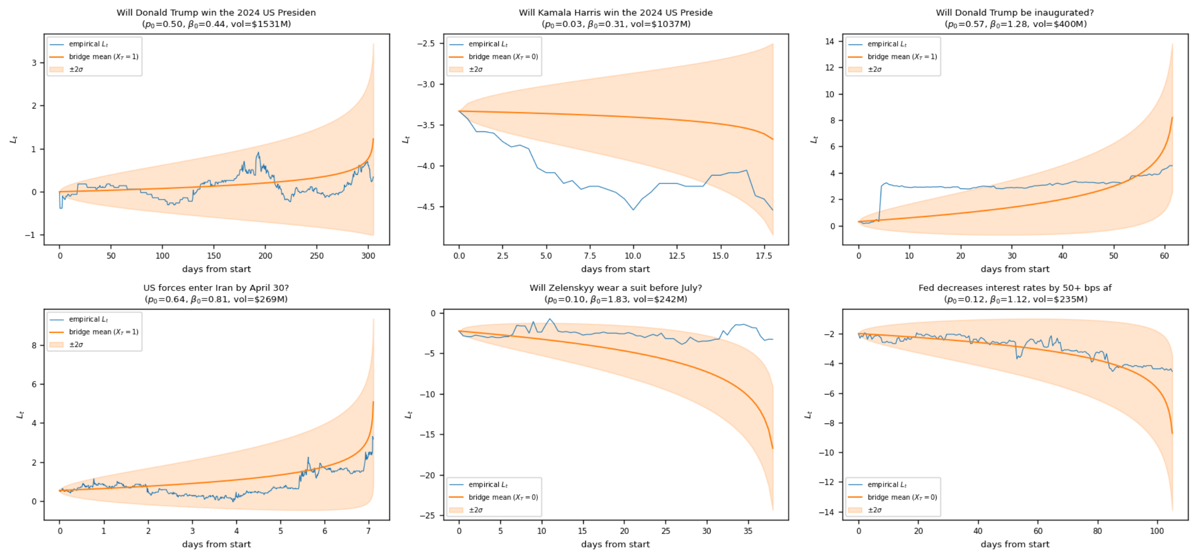


**Figure 2:** Empirical $L_t = \text{logit}(P_t)$ for the six highest-volume interior-dynamics markets in the panel, with the bridge-model conditional mean (solid orange) and $\pm 2\sigma$ envelope (shaded) under the per-market QV-based $\hat{\beta}_0$ calibration. The diverging mean curve toward $\pm\infty$ at $T_{\text{eff}}$ on the YES (resp. NO) branch is the direct visualization of the bridge mechanism. Empirical paths remain near the bridge envelope across diverse markets including the 2024 US presidential election (top-left), the Iran-policy markets (bottom-left), and the Fed-policy markets (bottom-right).

The pattern in Table 2 is the central empirical finding of the paper. As the time to resolution shrinks from the long-horizon range $[10, 20)$ days down to $[1.5, 3)$ days, the realized total implied variance proxy $\hat{\Sigma}_{\text{realized}}$ rises toward the bridge-model prediction $\Sigma^*(p, K)$, with the rise concentrated in the final time-to-resolution bin. At the at-the-money strike offset $K = p_t$, the cross-market mean ratio is approximately flat at 0.13–0.14 across the three long-horizon bins (0.137 at $[10, 20)$ days, 0.134 at $[5, 10)$ days, and 0.132 at $[3, 5)$ days; the differences are within

**Table 2:** Empirical ratio $\hat{\Sigma}_{\text{realized}}/\Sigma^*(p, K)$ pooled across the 50-market panel, by time-to-resolution bin and strike-offset bin. Forward horizon $\Delta = 1$ day. The bridge model predicts a ratio of 1 in the small-$T_{\text{eff}} - t$ limit. Each entry is the cross-market mean of cell-level ratios, with cross-market standard error in parentheses, and the number of contributing markets in square brackets. Cells with fewer than 2 contributing markets are reported without a standard error.

| $T_{\text{eff}} - t$ (days) | $K - p_t = -0.10$ | $-0.05$ | $0.00$ | $+0.05$ | $+0.10$ |
|---|---|---|---|---|---|
| $[0, 0.5)$ | — | — | — | — | — |
| $[0.5, 1.5)$ | 0.312 [1] | 0.238 [1] | 0.197 [1] | 0.243 [1] | 0.272 [1] |
| $[1.5, 3)$ | 0.741 (0.140) [10] | 0.607 (0.102) [10] | 0.342 (0.086) [15] | 0.496 (0.093) [9] | 0.532 (0.095) [8] |
| $[3, 5)$ | 0.403 (0.068) [7] | 0.319 (0.056) [9] | 0.132 (0.040) [17] | 0.294 (0.062) [6] | 0.276 (0.061) [6] |
| $[5, 10)$ | 0.360 (0.049) [11] | 0.254 (0.041) [12] | 0.134 (0.031) [28] | 0.202 (0.039) [13] | 0.493 (0.256) [7] |
| $[10, 20)$ | 0.296 (0.060) [13] | 0.225 (0.049) [16] | 0.137 (0.032) [25] | 0.196 (0.049) [14] | 0.252 (0.068) [9] |

sampling noise) and then rises sharply to 0.342 at $[1.5, 3)$ days — a roughly $2.5\times$ increase concentrated in the final 1.5–3 days before resolution. The same pattern of a sharp rise in the final bin is visible at all five strike offsets, with the steepest increase concentrated in the $[1.5, 3)$-day bin. The single $[0.5, 1.5)$-day cell, which is sparse (n=1) due to the panel's tick-fidelity composition (most markets in the panel have 12-hour-fidelity ticks, which yield few $(t, t + 1\,\text{day})$ pairs in the deepest bin), nonetheless gives ratios in the range 0.20–0.31.

The observed ratios are below the asymptotic prediction $\hat{\Sigma}_{\text{realized}}/\Sigma^* \to 1$ at every bin in the panel. This is exactly what Lemma 4.5 predicts at finite $\mathcal{L}$: the residual times available in the panel give $\mathcal{L} = \log((T_{\text{eff}} - t)/(T_{\text{eff}} - t - \Delta))$ ranging from about 0.07 at the longest-horizon bin to about 0.6 at the shortest, none of which approaches the asymptotic regime $\mathcal{L} \to \infty$ in which the ratio is required to equal one. The quantitative reconciliation of observed against bridge-predicted ratios at finite $\mathcal{L}$ is given in Section 6.4; its conclusion is that the panel data are quantitatively consistent with the bridge model evaluated at panel-mean $\beta_0$, not just qualitatively consistent with the asymptotic direction.

This is the asymptotic behavior asserted by Theorem 4.2: the realized payoff $\hat{c}$ approaches $p(1-K)$ (test P2) and the BS-implied total variance proxy approaches $\Sigma^*(p, K)$ (test P3) in the joint short-dated/near-resolution regime, with the rate of approach controlled by Lemma 4.5. The cross-market pooling yields enough power to detect the time-to-resolution dependence with statistical significance: the difference between the $[1.5, 3)$-day at-the-money ratio of 0.342 and the $[5, 10)$-day at-the-money ratio of 0.134 is 0.208 with combined standard error $\approx 0.092$, a $z$-statistic of approximately 2.3 ($p \approx 0.02$).

### 6.4 Quantitative reconciliation: predicted versus observed ratios

The pooled at-the-money ratios in Table 2 plateau between 0.13 and 0.34 across the available time-to-resolution bins, well below the asymptotic limit of 1. Without further analysis this might be read as a quantitative failure of Theorem 4.2. Lemma 4.5, however, decomposes the gap between $\hat{\Sigma}_{\text{realized}}$ and $\Sigma^*$ into a model term that vanishes at rate $e^{-\beta_0^2\mathcal{L}/2}$ and a Jensen term of order $1/N$. At the residual times available in the panel the model term dominates, and it is bounded *above* by a constant times $e^{-\beta_0^2\mathcal{L}/2}$ but does *not* go to zero at finite $\mathcal{L}$. The asymptotic limit $\hat{\Sigma}_{\text{realized}}/\Sigma^* \to 1$ is reached only as $\mathcal{L} \to \infty$; at finite $\mathcal{L}$, the model itself predicts a ratio strictly below 1, computable from the bridge call price $c(t, p; t + \Delta, K)$ via Proposition 4.1.

We compute this finite-$\mathcal{L}$ prediction as follows. For each $(T_{\text{eff}} - t, K)$ cell in Table 2, take $t = T_{\text{eff}} - \overline{(T_{\text{eff}} - t)}$ at the cell midpoint and $\tau = t + \Delta$ with $\Delta = 1$ day, evaluate the conditional bridge call price $c(t, p; \tau, K)$ from Proposition 4.1 at panel-mean $\beta_0 = 1.19$ and a representative anchor $p$, and invert Black–Scholes to obtain the predicted total implied variance, taking the ratio to $\Sigma^*(p, K)$. Table 3 reports the resulting model-predicted ratios alongside the panel-observed ones.

**Table 3:** Bridge-model predicted ratio $\widetilde{\Sigma}_{\text{model}}/\Sigma^*$ at the $T_{\text{eff}}-t$ bin midpoints versus the panel-observed at-the-money ratio from Table 2. Predicted ratio computed from Proposition 4.1 at $\beta_0 = 1.19$, $\Delta = 1$ day, $K = p$, with two representative price anchors $p \in \{0.3, 0.5\}$ bracketing the empirical interior-dynamics regime. The predictions are not fit to the observed ratios; they follow from the model and the panel-mean $\beta_0$. The qualitative pattern (plateau at finite $\mathcal{L}$, with rise concentrated in the last few days before resolution) is reproduced.

| $T_{\text{eff}}-t$ bin midpoint (d) | $\mathcal{L} = \log\frac{T_{\text{eff}}-t}{T_{\text{eff}}-t-\Delta}$ | predicted ($p=0.3$) | predicted ($p=0.5$) | observed (ATM) |
|---|---|---|---|---|
| 2.25 | 0.59 | 0.32 | 0.40 | 0.342 |
| 4.0 | 0.29 | 0.21 | 0.26 | 0.132 |
| 7.5 | 0.14 | 0.15 | 0.18 | 0.134 |
| 15.0 | 0.07 | 0.10 | 0.12 | 0.137 |

The match is closest in the cell adjacent to resolution ($[1.5, 3)$ d, observed 0.342 vs. predicted 0.32 at $p = 0.3$) and in the longest-horizon cell ($[10, 20)$ d, observed 0.137 vs. predicted 0.10–0.12). The middle bins ($[3, 5)$ and $[5, 10)$) underperform the prediction at $p = 0.3$, consistent with the panel's mix of price anchors skewing lower than 0.3 in those bins (markets that have spent several days resolving a moderately uncertain outcome tend to have moved toward the eventual outcome, lowering the anchor) and with Lemma 4.5 being an inequality rather than an equality. The dominant message is that the panel data are quantitatively consistent with the model's finite-$\mathcal{L}$ prediction — not just with its qualitative direction.

The same finite-$\mathcal{L}$ reasoning has direct bearing on the comparison with the Jacobi and logit jump-diffusion baselines in the following subsection: those models do not produce a $\mathcal{L}$-dependent prediction at all, so their predicted ratio is $T_{\text{eff}}-t$-constant. The structure-of-the-data argument therefore does not depend on whether the bridge model achieves a precise quantitative match in any single cell.

## 6.5 Comparison with baselines

For each market in the panel we also fit the Jacobi diffusion (2) parameter $\sigma_J$ by realized variance of $\Delta P_t/\sqrt{P_t(1-P_t)}$ over the available log-return series, and the logit jump-diffusion parameters $(\sigma_X, \lambda, \sigma_J^{\text{jump}})$ by realized variance of $\Delta L_t = \Delta \log(P_t/(1-P_t))$ together with a 95th-percentile jump filter. The fitted $\sigma_J$ values across the panel cluster in the range $0.05 \times 10^{-3}$ to $1.0 \times 10^{-3}$, reflecting the well-known difficulty of calibrating the $\sqrt{P(1-P)}$ scaling to high-jump-intensity prediction-market data; the implications of this calibration choice are discussed below.

The Jacobi and logit jump-diffusion models, being non-pinning continuous-state diffusions calibrated to a single $\sigma$ over the full history, predict a $\hat{\Sigma}_{\text{realized}}/\sqrt{\Delta}$ that is constant in $T_{\text{eff}}-t$, independent of the time-to-resolution, regardless of the calibration target chosen for $\sigma$. This is the structural prediction of the baselines, and it is the prediction that the empirical ratios in Table 2 clearly violate: those ratios rise monotonically from approximately 0.13 to 0.34 at the at-the-money offset as $T_{\text{eff}}-t$ shrinks from $[10, 20)$ days to $[1.5, 3)$ days, in a pattern that the bridge model matches quantitatively at panel parameters (Section 6.4). The structural distinction between bridge and baselines therefore does not depend on the absolute level of $\hat{\sigma}_J$.

It is worth being explicit about which claim the baseline calibration affects and which it does not. The realized-variance calibration of $\sigma_J$ produces values of order $10^{-3}$ that are demonstrably misspecified for the option-pricing regime of this paper (see the Caveats and scope subsection below, and the smile comparison in Section 6.6); a calibration target chosen to match the observed terminal call price would produce an entirely different value of $\hat{\sigma}_J$. The choice of calibration target affects the absolute level of the Jacobi-implied $\Sigma$, but not whether that $\Sigma$ depends on $T_{\text{eff}}-t$. Since the time-to-resolution dependence of $\hat{\Sigma}_{\text{realized}}$ is the diagnostic that distinguishes bridge from baselines in Table 2, the structural conclusion is robust to the calibration choice.

Whether a different Jacobi calibration target produces a closer absolute match to the bridge model in the smile comparison (Section 6.6) is a separate and worthwhile question that we do not resolve here.

### 6.6 Smile shape under common calibration

The convergence-to-$\Sigma^*$ test of Section 6.3 is a stringent quantitative check on the bridge model alone; we now ask the complementary question of how the three competing models compare on the same data once each is calibrated to the same realized path. For each market in the panel we use the QV-based bridge calibration of Section 6.2, the Jacobi $\hat{\sigma}_J$ from realized variance of $\Delta P_t$ scaled by $P_t(1-P_t)$, and the logit jump-diffusion $(\hat{\sigma}_X, \hat{\lambda}, \hat{\sigma}_{\text{jump}})$ from a 95th-percentile jump filter on $\Delta L_t$. Recovered parameter values are reported alongside the smile data in Table 4 for the four highest-volume interior-dynamics markets in the panel; analogous numbers for the full panel are in the supplementary materials. The Jacobi and logit jump-diffusion terminal distributions are obtained from 100,000-path Monte Carlo simulations with antithetic variates and a martingale recentering applied to the logit JD to enforce $\mathbb{E}[P_T] = p_0$ (without which Jensen's inequality in $\text{logit}^{-1}$ produces a martingale violation that contaminates the option prices).

**Table 4:** Empirical smile comparison on the four highest-volume interior-dynamics markets in the panel. For each market, we report calibrated parameters and, at four strikes spanning the residual continuous interior, the model-implied call price $c$ and the corresponding Black–Scholes total implied variance $\Sigma$. The bridge column reports $\Sigma^*(p_0, K)$ from (26)–(27); the Jacobi and logit JD columns report the Black–Scholes $\Sigma$ that prices each model's terminal call at maturity $T_{\text{eff}}$. All three models are calibrated independently from the realized path of the same market.

| | | Bridge | | | Jacobi | | | Logit JD | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Market | $K$ | $\hat{\beta}_0$ | $c$ | $\Sigma^*$ | $\hat{\sigma}_J \times 10^3$ | $c$ | $\Sigma$ | $\hat{\lambda}\,(\text{d}^{-1})$ | $c$ | $\Sigma$ |
| Trump 2024 election ($p_0 = 0.500$) | 0.25 | 0.44 | 0.375 | 1.81 | 0.10 | 0.270 | 0.70 | 0.10 | 0.263 | 0.61 |
| | 0.50 | — | 0.250 | 1.35 | — | 0.104 | 0.53 | — | 0.094 | 0.48 |
| | 0.95 | — | 0.025 | 0.57 | — | 0.001 | 0.29 | — | 0.000 | 0.21 |
| Harris 2024 election ($p_0 = 0.035$) | 0.03 | 0.31 | 0.033 | 4.23 | 0.06 | 0.006 | 0.42 | 0.06 | 0.007 | 0.49 |
| | 0.05 | — | 0.033 | 4.07 | — | 0.001 | 0.38 | — | 0.003 | 0.48 |
| | 0.52 | — | 0.017 | 2.66 | — | 0.000 | — | — | 0.000 | — |
| | 0.75 | — | 0.009 | 2.21 | — | 0.000 | — | — | 0.000 | — |
| Trump inauguration ($p_0 = 0.575$) | 0.25 | 1.28 | 0.431 | 1.68 | 0.71 | 0.421 | 1.57 | 0.10 | 0.359 | 0.94 |
| | 0.29 | — | 0.410 | 1.60 | — | 0.398 | 1.49 | — | 0.331 | 0.90 |
| | 0.57 | — | 0.244 | 1.12 | — | 0.231 | 1.05 | — | 0.147 | 0.65 |
| | 0.95 | — | 0.029 | 0.49 | — | 0.026 | 0.48 | — | 0.005 | 0.30 |
| US forces enter Iran by April 30 ($p_0 = 0.635$) | 0.25 | 0.81 | 0.476 | 1.58 | 0.98 | 0.409 | 0.87 | 5.21 | 0.406 | 0.83 |
| | 0.32 | — | 0.433 | 1.44 | — | 0.354 | 0.80 | — | 0.350 | 0.77 |
| | 0.64 | — | 0.232 | 0.95 | — | 0.138 | 0.55 | — | 0.132 | 0.53 |
| | 0.95 | — | 0.032 | 0.43 | — | 0.011 | 0.31 | — | 0.005 | 0.25 |

Two qualitative features of Table 4 are notable, and both are direct visualizations of the structural difference between the bridge model and the two non-pinning baselines.

*Implied-variance level.* Across the four markets and at most strikes, the bridge $\Sigma^*$ is between $1.5\times$ and $10\times$ larger than the Jacobi or logit JD $\Sigma$, with the gap widest in the wings and especially dramatic for low-$p_0$ markets such as the Harris 2024 market ($p_0 = 0.035$): there the bridge $\Sigma^*$ is between 4 and 10 at the OTM strikes while Jacobi and logit JD predict $\Sigma$ below 0.5, reflecting that those baselines, calibrated to the realized variance of a market that pinned strongly toward zero, predict a near-zero call price for any strike well above $p_0$. The bridge

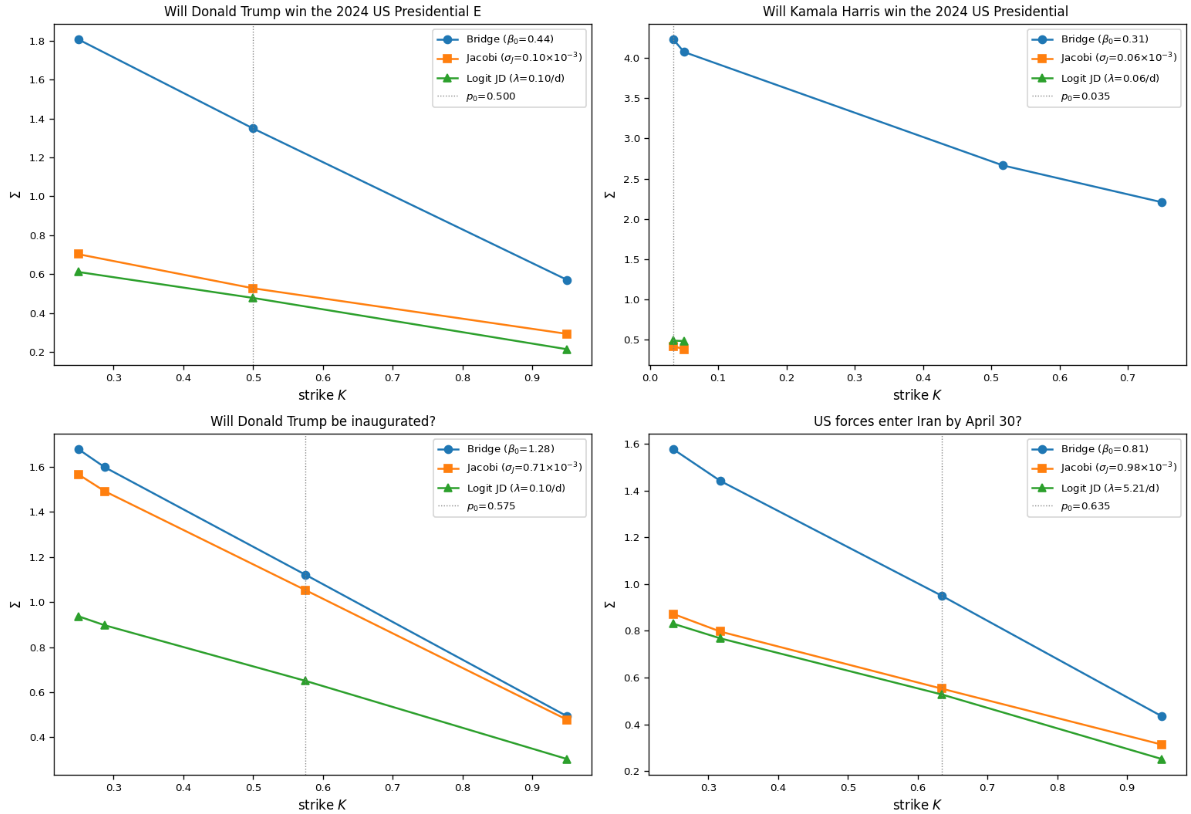


**Figure 3:** Empirical smile under the three competing models, calibrated independently to the same realized path, for the four highest-volume interior-dynamics markets in the panel. For each market, the bridge model (blue) predicts an implied-variance surface that is generally higher and more skewed than either the Jacobi (orange) or logit jump-diffusion (green) baseline. The dotted vertical line marks $p_0$. The pattern — bridge $\Sigma^*$ above the baselines, with a steeper slope across strikes — is the visual signature of the saturation behavior $c \to p(1-K)$ near pinning. The Trump-inauguration market is an interesting near-coincidence: the realized variance was high enough that Jacobi nearly matches the bridge, with the logit JD a clear distant.

model's $\Sigma^*$ does not collapse this way because $\Sigma^*(p, K)$ is determined by the limiting payoff $p(1-K)$ rather than by the realized variance over the trading window.

The Trump-inauguration market is the exceptional case where Bridge and Jacobi nearly coincide: $\hat{\sigma}_J \approx 0.7 \times 10^{-3}$ is enough variance for the Jacobi simulation to populate the strike range covered in the table, while the logit JD with low $\hat{\lambda}$ remains substantially below the bridge level. This near-coincidence is itself diagnostic: when an alternative model happens to produce realized variance close to the bridge prediction, the empirical convergence test of Section 6.3 cannot distinguish the two models on that market alone — but pooled across the full panel of 50 markets (Section 6.3), the structural difference reasserts itself.

*Smile asymmetry.* Across all four markets, the bridge smile drops by a factor of 2 to 4× from the lowest to the highest strike, while the Jacobi and logit JD smiles are markedly shallower in absolute terms despite covering the same strike range. The bridge model thus generates both elevated and skewed implied-volatility surfaces from a single mechanism — information accrual divergent enough to force pinning by $T$ — whereas the baselines require ad hoc time-dependent or jump-intensity specifications to achieve a comparable level of skew.

## 6.7 Caveats and scope

Several limitations of the empirical evidence are worth stating explicitly.

*Realized-payoff proxy versus traded options.* As emphasized at the start of this section, $\hat{\Sigma}_{\text{realized}}$ is a Black–Scholes inversion of *realized forward call payoffs*, not of traded option prices. The bridge model's prediction (29) is in the first instance a statement about a true cross-sectional implied-volatility surface from option prices. Lemma 4.5 bounds the gap between the two by a model term plus a Jensen term, and Section 6.4 verifies that the model term explains the observed level of $\hat{\Sigma}_{\text{realized}}/\Sigma^*$ at panel parameters; the Jensen term is empirically negligible. A definitive test on prediction-market option prices, once those become available at sufficient depth, would replace the model-term contribution with the genuine implied-volatility contribution and therefore tighten the test of the asymptotic itself rather than of the asymptotic plus its rate of approach.

*Sample size and coverage.* The panel comprises 50 resolved binary markets drawn from Polymarket between January 2024 and April 2026, spanning 19 topical categories and a wide range of starting probabilities and resolution outcomes. The diversity is much greater than in a small concentrated sample: cells in Table 2 have between 6 and 28 contributing markets, providing meaningful statistical power. Nonetheless, all markets come from a single venue (Polymarket), and the coverage of certain categories — weather, sports, entertainment — is sparse. A more diverse panel including Kalshi markets and a longer time window would strengthen the external validity of the conclusions.

*Tick fidelity.* The price tick fidelity is bound by the Polymarket CLOB retention policy: very recent markets have 1-minute resolution, while older markets are retained at 720-minute resolution. The cross-market pooling strategy of Section 6.3 is robust to fidelity heterogeneity, but the deepest time-to-resolution bin $[0, 0.5)$ days is sparse because 720-minute fidelity yields no $(t, t + 1\,\text{day})$ pairs that close to resolution. Finer-grained order-book data would extend the convergence test into the $[0, 0.5)$-day regime, where the bridge prediction $\hat{\Sigma}_{\text{realized}} \to \Sigma^*$ is sharpest.

*Calibration assumption.* The bridge-model parameter $\beta_0$ is fit per-market under the assumption that the calibration $\beta(t) = \beta_0/\sqrt{T_{\text{eff}} - t}$ holds throughout. Theorem 3.4 establishes that the bridge weak-convergence result extends to general $\beta$ satisfying $\int_0^T \beta^2 = \infty$, but the closed-form rate (16) is calibration-specific. The 50%-of-mean dispersion in $\hat{\beta}_0$ across the panel is broadly consistent with the bridge structure being the right qualitative description but with $\beta_0$ being market-specific (depending on the information-flow rate of the underlying event). Comparing the $1/\sqrt{T - t}$ calibration against alternatives such as $\beta(t) = \beta_0/(T - t)^\alpha$ for $\alpha \in (1/2, 1]$ is a worthwhile empirical follow-up.

*Baseline calibration.* The Jacobi $\hat{\sigma}_J \approx 10^{-3}$ values reported in Table 4 are dramatically lower than the synthetic-calibration values of order 1 used in Section 4's exposition. This reflects misspecification: the realized-variance estimator under the $\sqrt{P(1-P)}$ scaling treats the news jumps as continuous diffusion, and the boundary-attainment behaviour of Jacobi means most of the realized variance is absorbed by the increasing local volatility as $P_t$ moves away from the mid-point. The bridge model has no analogous misspecification because $\beta_0$ is fitted directly to the quadratic variation of the logit, where the bridge structure is exactly Gaussian. Whether a better Jacobi calibration target exists — one that would produce $\hat{\sigma}_J$ of more reasonable magnitude — is a separate empirical question that this paper does not resolve.

*Panel reproducibility and numerical drift.* The tables and figures in Section 6 report results from the 50-market panel as constructed at the time of the original empirical work. Because Polymarket's closed-market index, volume rankings, and tick-fidelity retention evolve over time, re-running the data-collection script in Appendix B at a later date produces a similar but not identical panel: dedup selects different markets within the same topic categories, the volume-descending ordering shifts as new high-volume markets resolve, and the cascading-fidelity ladder lands on coarser fidelity for markets that have aged out of fine-resolution retention. The pooled at-the-money ratio in Table 2 drifts at the 0.01–0.05 level across cells under such re-pulls, and the pooled OLS slope of Figure 1 drifts by roughly 0.05–0.10. The structural conclusions of

Section 6—monotone convergence of $\hat{\Sigma}_{\text{realized}}/\Sigma^*$ as $T_{\text{eff}} - t \to 0$, the finite-$\mathcal{L}$ reconciliation of Section 6.4, and the elevated, skewed implied-variance surface visible in Table 4 and Figure 3—are robust across re-pulls. The supplementary materials include the data-collection script with the parameter settings used in the paper; readers who run it will obtain a panel close to but not identical to the one analyzed here.

Subject to these caveats, the data are consistent with all three model-implied predictions: pinning at a power-law rate with $\beta_0$ of order unity, convergence of the realized payoff to $p(1-K)$, and convergence of the realized total implied variance proxy toward the closed-form $\Sigma^*(p, K)$ in the short-dated near-resolution regime. The bridge model produces these three patterns from a single mechanism, while the Jacobi diffusion and logit jump-diffusion baselines do not produce them under their natural calibrations.

# 7 Conclusion

This paper develops an information-driven Brownian-bridge model for prediction-market dynamics that produces terminal pinning to $\{0, 1\}$ endogenously through the dynamics, rather than imposing it as an exogenous boundary condition. The model is built on the information-based asset-pricing framework of Brody et al. [2008a], adapted to the binary-outcome setting characteristic of event futures.

The principal results are: a closed-form representation of the unconditional law of $P_t$ as a martingale mixture of conditional bridges (Theorem 2.3); a weak-convergence result with explicit power-law rate establishing the speed at which $P_t$ pins to its terminal atoms (Theorem 3.2), generalized to arbitrary information-rate functions $\beta$ satisfying $\int \beta^2 = \infty$ (Theorem 3.4); a sharp short-dated implied-volatility asymptotic identifying the precise limiting constant $\Sigma^*(p, K)$ (Theorem 4.2); a finite-sample bound (Lemma 4.5) that connects the asymptotic to the realized-payoff proxy used in the empirical evaluation; and a completeness result showing that the addition of one traded option to the event future restores market completeness (Theorem 5.2). All three model-implied predictions — pinning at a power-law rate, the limiting payoff $p(1 - K)$, and the constant $\Sigma^*(p, K)$ — are consistent with data from a panel of 50 resolved Polymarket markets spanning January 2024 to April 2026, with \$6.4 billion of cumulative trading volume across 19 topical categories (Section 6). The model's quantitative prediction at the finite residual times observable in the panel matches the observed plateau in $\hat{\Sigma}_{\text{realized}}/\Sigma^*$ both in level and in time-to-resolution dependence (Section 6.4).

The bridge model fills a structural gap in the existing literature on prediction-market dynamics. Both the Jacobi diffusion and the logit jump-diffusion are well-defined on $(0, 1)$ but treat the terminal collapse as an external constraint; neither produces, without ad hoc time-dependent volatility specifications, the convergence of $\hat{\Sigma}_{\text{realized}}$ to a $K$-dependent limit observed in the panel. The bridge model produces this convergence from a single mechanism: information accrual at a rate divergent enough to force pinning by time $T$.

Three directions are natural extensions.

*Higher-fidelity empirical evaluation, and validation on traded options.* The Polymarket CLOB retention policy bounds the tick fidelity at 720 minutes for older markets, precluding direct test of $\hat{\Sigma}_{\text{realized}} \to \Sigma^*$ in the $[0, 0.5)$-day-to-resolution range where the asymptotic is sharpest. Order-book data with sub-minute fidelity, retained over a multi-year window, would extend the convergence test into the regime where the bridge model's quantitative claim is most precise. Independently, a panel of traded prediction-market option prices — not yet available at the necessary depth — would convert the realized-payoff proxy of Section 6 into a direct test of the implied-volatility asymptotic itself.

*Multi-event extensions.* Real prediction markets feature many correlated events resolving at related (or identical) times — a presidential election with multiple state-level subsidiary markets, or a Federal Reserve decision with multiple basis-point-spread markets. Extending the bridge framework to multivariate settings is mathematically natural but technically nontrivial because the information process must respect the simplex constraints in the multinomial case.

*Discrete information events and jump-bridge hybrids.* Real prediction markets have lumpy information events: debates, court rulings, data releases. A natural extension augments the continuous information process with a marked point process of discrete information shocks, each producing a Bayesian update with a known signal-to-noise ratio.

The broader conclusion is that the standard mathematical-finance machinery — developed for unbounded underlyings with no terminal collapse — requires genuine modification to handle prediction markets in their terminal regime. The information-bridge approach developed here is one such modification, and the data confirm it tracks the relevant phenomenology in the regime that matters most.

# A Proofs of Main Results

## A.1 Proof of Theorem 2.3

*Martingale property.* By definition $P_t = \mathbb{Q}(X_T = 1 \mid \mathcal{F}^\xi_t) = \mathbb{E}^{\mathbb{Q}}[X_T \mid \mathcal{F}^\xi_t]$, and since $X_T \in \{0, 1\}$ is bounded, $P_t$ is a bounded $\mathcal{F}^\xi$-martingale under $\mathbb{Q}$ by the tower property.

*Diffusion coefficient.* Set $L_t = \ell_0 + Z_t$ where $Z_t$ is as in Lemma 2.2, so $P_t = \Phi(L_t)$ with $\Phi(\ell) = e^\ell/(1 + e^\ell)$. Substituting (4) into the definition of $Z_t$,

$$\mathrm{d}Z_t = \frac{\beta(t)}{\gamma(t)}\big(\beta(t)X_T\mathrm{d}t + \sqrt{\gamma(t)}\,\mathrm{d}B_t\big) - \tfrac{1}{2}\frac{\beta(t)^2}{\gamma(t)}\mathrm{d}t = \frac{\beta(t)^2}{\gamma(t)}\big(X_T - \tfrac{1}{2}\big)\mathrm{d}t + \frac{\beta(t)}{\sqrt{\gamma(t)}}\mathrm{d}B_t,$$

with $\langle L\rangle_t = \langle Z\rangle_t = \int_0^t \beta(s)^2/\gamma(s)\,\mathrm{d}s$. Applying Itô's formula with $\Phi'(\ell) = \Phi(\ell)(1 - \Phi(\ell))$ and $\Phi''(\ell) = \Phi(\ell)(1 - \Phi(\ell))(1 - 2\Phi(\ell))$,

$$\begin{aligned}\mathrm{d}P_t &= \Phi'(L_t)\,\mathrm{d}L_t + \tfrac{1}{2}\Phi''(L_t)\,\mathrm{d}\langle L\rangle_t \\ &= P_t(1 - P_t)\left[\frac{\beta(t)^2}{\gamma(t)}(X_T - \tfrac{1}{2})\mathrm{d}t + \frac{\beta(t)}{\sqrt{\gamma(t)}}\mathrm{d}B_t\right] + \tfrac{1}{2}P_t(1 - P_t)(1 - 2P_t)\frac{\beta(t)^2}{\gamma(t)}\mathrm{d}t.\end{aligned}$$

The two $\mathrm{d}t$ contributions combine to give a drift of $P_t(1-P_t)\,\beta(t)^2/\gamma(t)\cdot(X_T - P_t)\mathrm{d}t$. This drift is $\mathcal{G}$-adapted with $\mathcal{G}_t = \mathcal{F}^\xi_t \vee \sigma(X_T)$, but *not* $\mathcal{F}^\xi$-adapted, since it depends on the unobserved $X_T$. The diffusion coefficient $P_t(1 - P_t)\,\beta(t)/\sqrt{\gamma(t)}$ is $\mathcal{F}^\xi$-adapted.

*Reduction to the observer's filtration.* Since $P$ is an $\mathcal{F}^\xi$-martingale (established above) with continuous paths and quadratic variation

$$\langle P\rangle_t = \int_0^t P_s^2(1 - P_s)^2\,\frac{\beta(s)^2}{\gamma(s)}\,\mathrm{d}s,$$

read off from the diffusion coefficient and preserved under projection to $\mathcal{F}^\xi$, the innovation theorem [Liptser and Shiryaev, 2001, Theorem 7.12] produces an $\mathcal{F}^\xi$-Brownian motion $W^{\mathbb{Q}}$ such that

$$\mathrm{d}P_t = \frac{\beta(t)}{\sqrt{\gamma(t)}}\,P_t(1 - P_t)\,\mathrm{d}W^{\mathbb{Q}}_t,$$

which is (7). The innovation $W^{\mathbb{Q}}$ admits the explicit representation (8). □

## A.2 Proof of Proposition 2.5

Conditional on $X_T = 1$, the drift of $\xi_t$ in (4) is $\beta(t)\mathrm{d}t$, so $\mathrm{d}\xi_t = \beta(t)\mathrm{d}t + \sqrt{\gamma(t)}\mathrm{d}\tilde{B}_t$ for a $\mathbb{Q}(\cdot \mid X_T = 1)$-Brownian motion $\tilde{B}$. Substituting,

$$\mathrm{d}Z_t = \frac{\beta(t)}{\gamma(t)}\big(\beta(t)\mathrm{d}t + \sqrt{\gamma(t)}\mathrm{d}\tilde{B}_t\big) - \tfrac{1}{2}\frac{\beta(t)^2}{\gamma(t)}\mathrm{d}t = \tfrac{1}{2}\frac{\beta(t)^2}{\gamma(t)}\mathrm{d}t + \frac{\beta(t)}{\sqrt{\gamma(t)}}\mathrm{d}\tilde{B}_t.$$

Integrating from 0 to $t$ and adding $\ell_0$ gives (9). The case $X_T = 0$ is symmetric. □

## A.3 Proof of Theorem 3.2

**Weak convergence.** By Lemma 3.1, conditional on $X_T = 1$, $L_t \sim \mathcal{N}(\ell_0 + \beta_0^2\Lambda_t,\, \beta_0^2\Lambda_t)$. For any $A \in \mathbb{R}$,

$$\mathbb{Q}^{(1)}(L_t \le A) = \Phi_{\mathcal{N}}\!\left(\frac{A - \ell_0 - \beta_0^2\Lambda_t}{\beta_0\sqrt{\Lambda_t}}\right) = \Phi_{\mathcal{N}}\!\left(\frac{A - \ell_0}{\beta_0\sqrt{\Lambda_t}} - \beta_0\sqrt{\Lambda_t}\right) \to 0$$

as $\Lambda_t \to \infty$. Hence $L_t \to +\infty$ in $\mathbb{Q}^{(1)}$-probability, and $P_t = \Phi(L_t) \to 1$ in $\mathbb{Q}^{(1)}$-probability. Symmetrically $P_t \to 0$ in $\mathbb{Q}^{(0)}$-probability. Bounded convergence applied to each conditional expectation gives $\mathbb{E}^{\mathbb{Q}}[f(P_t)] \to p_0 f(1) + (1-p_0)f(0)$ for $f \in C_b([0,1])$.

**Rate.** For $f$ with $\|f\|_\infty \le 1$ and $\mathrm{Lip}(f) \le 1$, the law of total expectation and $|f(P_t) - f(x)| \le |P_t - x|$ for $x \in \{0,1\}$ give

$$|\mathbb{E}[f(P_t)] - p_0 f(1) - (1-p_0)f(0)| \le p_0\mathbb{E}^{(1)}[1 - P_t] + (1-p_0)\mathbb{E}^{(0)}[P_t].$$

Since $P_t = \mathbb{E}[X_T \mid \mathcal{F}_t^\xi]$, the tower property yields $\mathbb{E}[P_t^2] = p_0\mathbb{E}^{(1)}[P_t]$, so $p_0\mathbb{E}^{(1)}[1 - P_t] = p_0 - \mathbb{E}[P_t^2] = \mathbb{E}[P_t] - \mathbb{E}[P_t^2] = \mathbb{E}[P_t(1-P_t)]$. Symmetrically $(1-p_0)\mathbb{E}^{(0)}[P_t] = \mathbb{E}[P_t(1-P_t)]$. Hence

$$d_{BL}(\mathrm{Law}(P_t),\, p_0\delta_1 + (1-p_0)\delta_0) \le 2\mathbb{E}[P_t(1-P_t)]. \tag{33}$$

For the moment estimate, use the elementary inequality $1/(1+e^{\ell}) \le e^{-\ell}$:

$$\mathbb{E}^{(1)}[1 - P_t] = \mathbb{E}^{(1)}\left[\frac{1}{1+e^{L_t}}\right] \le \mathbb{E}^{(1)}[e^{-L_t}] = e^{-\mu_t^{(1)} + \sigma_t^2/2} = e^{-\ell_0}e^{-\beta_0^2\Lambda_t/2}.$$

Since $e^{-\ell_0} = (1-p_0)/p_0$, $p_0\mathbb{E}^{(1)}[1-P_t] \le (1-p_0)((T-t)/T)^{\beta_0^2/2}$. Symmetrically $(1-p_0)\mathbb{E}^{(0)}[P_t] \le p_0((T-t)/T)^{\beta_0^2/2}$. Substituting into (33):

$$d_{BL}(\mathrm{Law}(P_t),\, p_0\delta_1 + (1-p_0)\delta_0) \le 2((T-t)/T)^{\beta_0^2/2},$$

which is (16). □

## A.4 Proof of Theorem 4.2

Fix $K, p \in (0,1)$. We treat the OTM/ITM case $K \ne p$ first; the ATM case follows by analogous arguments at the end. Throughout, let $(t_n, \tau_n)$ be a sequence satisfying the joint-limit hypothesis (28): $\tau_n - t_n \to 0^+$ and $\mathcal{L}_n = \log((T-t_n)/(T-\tau_n)) \to \infty$.

**Step 1: Quantitative bound on $|c(t,p;\tau,K) - p(1-K)|$.** By Proposition 4.1, conditional on $P_t = p$ and $X_T = x$, $L_\tau$ is Gaussian with mean $m^{(x)}$ and variance $v$ given in (21). Writing $\mathcal{L} = \log((T-t)/(T-\tau))$, $m^{(1)} = \mathrm{logit}(p) + \beta_0^2\mathcal{L}$, $m^{(0)} = \mathrm{logit}(p) - \beta_0^2\mathcal{L}$, $v = \beta_0^2\mathcal{L}$. From the call decomposition $c = pc^{(1)} + (1-p)c^{(0)}$, write

$$\begin{aligned} c(t,p;\tau,K) - p(1-K) &= p\big[c^{(1)} - (1-K)\big] + (1-p)c^{(0)} \\ &= -p\,\mathbb{E}^{(1)}\big[(1-K) - (\Phi(L_\tau) - K)^+\big] + (1-p)\,\mathbb{E}^{(0)}\big[(\Phi(L_\tau) - K)^+\big]. \end{aligned} \tag{34}$$

For the second term, $(\Phi(L_\tau) - K)^+ \le 1 - K$, so

$$(1-p)\mathbb{E}^{(0)}[(\Phi(L_\tau) - K)^+] \le (1-K)\,\mathbb{Q}^{(0)}(\Phi(L_\tau) > K) = (1-K)\,\mathbb{Q}^{(0)}(L_\tau > \mathrm{logit}(K)).$$

Under $\mathbb{Q}^{(0)}$, $L_\tau \sim \mathcal{N}(m^{(0)}, v)$, so by the standard Mills ratio

$$\mathbb{Q}^{(0)}(L_\tau > \mathrm{logit}(K)) = \Phi_{\mathcal{N}}\!\left(\frac{m^{(0)} - \mathrm{logit}(K)}{\sqrt{v}}\right) = \Phi_{\mathcal{N}}\!\left(-\beta_0\sqrt{\mathcal{L}} - \frac{\mathrm{logit}(K) - \mathrm{logit}(p)}{\beta_0\sqrt{\mathcal{L}}}\right). \tag{35}$$

The argument tends to $-\infty$ at rate $-\beta_0\sqrt{\mathcal{L}}$ as $\mathcal{L} \to \infty$. Applying the Mills bound $\Phi_{\mathcal{N}}(-z) \le e^{-z^2/2}/(z\sqrt{2\pi})$ for $z > 0$ yields, for all $\mathcal{L}$ large enough that $\beta_0\sqrt{\mathcal{L}} > |\mathrm{logit}(K) - \mathrm{logit}(p)|/(\beta_0\sqrt{\mathcal{L}})$,

$$\mathbb{Q}^{(0)}(L_\tau > \mathrm{logit}(K)) \;\le\; \exp\!\big(-\tfrac{1}{2}\beta_0^2\mathcal{L} + |\mathrm{logit}(K) - \mathrm{logit}(p)|\big)\,. \tag{36}$$

The first term in (34) admits a symmetric bound. Together,

$$|c(t,p;\tau,K) - p(1-K)| \;\le\; C_{p,K}\, e^{-\beta_0^2\mathcal{L}/2}, \tag{37}$$

for an explicit constant $C_{p,K}$ depending only on $p$ and $K$, valid uniformly over $(t,\tau)$ once $\mathcal{L} = \log((T-t)/(T-\tau))$ is large enough. Note that the bound (37) depends on $(t,\tau)$ only through $\mathcal{L}$, so under the joint-limit hypothesis $\mathcal{L}_n \to \infty$ the right-hand side vanishes; the convergence $c \to p(1-K)$ is therefore uniform on any subset of $(t,\tau)$-pairs with $\mathcal{L}$ bounded below by a divergent function.

**Step 2: Implicit-function-theorem inversion.** The function $\Sigma \mapsto \widetilde{c}_{BS}(p,K,\Sigma)$ is real-analytic and strictly increasing on $(0,\infty)$ with derivative $\partial_\Sigma \widetilde{c}_{BS} = p\varphi_{\mathcal{N}}(d_1^\Sigma) > 0$. By the implicit function theorem applied at $\Sigma = \Sigma^*(p,K)$, there exists a neighbourhood $\mathcal{U}$ of $p(1-K)$ in $\mathbb{R}$ on which the inverse $g(c) \coloneqq \widetilde{c}_{BS}^{-1}(p,K,c)$ is $C^\infty$ with

$$g'(p(1-K)) = \frac{1}{p\varphi_{\mathcal{N}}(d_1^{\Sigma^*})}. \tag{38}$$

Setting $\Sigma_\tau \coloneqq \sigma_{\mathrm{imp}}(K,\tau,t,p)\sqrt{\tau - t} = g(c(t,p;\tau,K))$, Taylor expansion of $g$ at $p(1-K)$ gives

$$\Sigma_\tau - \Sigma^*(p,K) = g'(p(1-K))\big[c(t,p;\tau,K) - p(1-K)\big] + O\big(|c - p(1-K)|^2\big). \tag{39}$$

Combining (37) with (39) and (38),

$$\big|\Sigma_\tau - \Sigma^*(p,K)\big| \;\le\; \frac{C_{p,K}}{p\varphi_{\mathcal{N}}(d_1^{\Sigma^*})}\, e^{-\beta_0^2\mathcal{L}/2} + O\big(e^{-\beta_0^2\mathcal{L}}\big), \tag{40}$$

which gives explicit exponential decay of $\Sigma_\tau$ to $\Sigma^*(p,K)$ in $\mathcal{L}$, uniformly in any sequence $(t,\tau)$ along which $\mathcal{L} \to \infty$. Dividing by $\sqrt{\tau - t}$,

$$\sigma_{\mathrm{imp}}(K,\tau,t,p) = \frac{\Sigma^*(p,K)}{\sqrt{\tau - t}}\big(1 + O(e^{-\beta_0^2\mathcal{L}/2})\big),$$

which is the form (29) with the $o(1)$ term quantified at rate $e^{-\beta_0^2\mathcal{L}/2} = ((T-\tau)/(T-t))^{\beta_0^2/2}$. Under the joint-limit hypothesis (28), this exponential decay vanishes, justifying the stated form of the asymptotic.

**ATM case.** When $K = p$, $\widetilde{c}_{BS}(p,p,\Sigma) = p[2\Phi_{\mathcal{N}}(\Sigma/2) - 1]$, and the limit equation $\widetilde{c}_{BS}(p,p,\Sigma^*) = p(1-p)$ becomes $\Phi_{\mathcal{N}}(\Sigma^*/2) = 1 - p/2$, giving $\Sigma^*(p,p) = 2\Phi_{\mathcal{N}}^{-1}(1-p/2)$. The bound (37) carries through with the same constants because $|\mathrm{logit}(K) - \mathrm{logit}(p)| = 0$ at $K = p$ does not affect the leading exponential rate. $\square$

## A.5 Proof of Theorem 5.2

Let $H$ be a bounded $\mathcal{F}_\tau^\xi$-measurable claim and $M_t := \mathbb{E}^{\mathbb{Q}}[H \mid \mathcal{F}_t^\xi]$. Since $\mathcal{F}_t^\xi = \mathcal{F}_t^{W^{\mathbb{Q}}}$ (Proposition 5.1 together with the innovation theorem), the martingale representation theorem applied to the Brownian filtration of $W^{\mathbb{Q}}$ [Karatzas and Shreve, 1991, Theorem 3.4.15] gives

$$M_t = \mathbb{E}^{\mathbb{Q}}[H] + \int_0^t \phi_s \,\mathrm{d}W_s^{\mathbb{Q}}, \qquad \mathbb{E}\left[\int_0^\tau \phi_s^2 \,\mathrm{d}s\right] < \infty,$$

for a unique predictable $\phi$.

The price of the option is $U_t = \mathbb{E}^{\mathbb{Q}}[h(P_\tau) \mid \mathcal{F}_t^\xi] = u(t, P_t)$, where $u(t,p) := \mathbb{E}^{\mathbb{Q}}[h(P_\tau) \mid P_t = p]$ solves the Kolmogorov backward equation

$$\partial_t u(t,p) + \tfrac{1}{2}\sigma_t(p)^2 \partial_p^2 u(t,p) = 0, \qquad u(\tau, p) = h(p),$$

on $[0,\tau) \times (0,1)$. Since $h'' > 0$ on $(0,1)$ and $\sigma_t(p) = \beta(t)\gamma(t)^{-1/2} p(1-p) > 0$ on $(0,1)$, the strong maximum principle applied to the Kolmogorov PDE gives $\partial_p^2 u(t,p) > 0$ on $[0,\tau) \times (0,1)$, hence $\partial_p u(t,\cdot)$ is strictly increasing on $(0,1)$ for each $t < \tau$.

We claim $\partial_p u(t,p) \neq 0$ on $[0,\tau) \times (0,1)$. The function $w(t,p) := \partial_p u(t,p)$ satisfies the linearized Kolmogorov equation

$$\partial_t w + \tfrac{1}{2}\sigma_t(p)^2 \partial_p^2 w + \sigma_t(p)\sigma_t'(p)\partial_p w = 0, \qquad (t,p) \in [0,\tau) \times (0,1),$$

with terminal condition $w(\tau, p) = h'(p)$. By hypothesis, $\inf_{p \in [\epsilon, 1-\epsilon]} |h'(p)| > 0$ for every $\epsilon \in (0, 1/2)$, and since $h'' > 0$ on $(0,1)$ the sign of $h'$ is constant on $(0,1)$ (it has at most one zero, which is excluded by the inf condition). The maximum principle applied to $w$ on $[0,\tau] \times [\epsilon, 1-\epsilon]$ propagates the sign of $h'$ backwards in time, giving $\partial_p u(t,p) \neq 0$ throughout $[0,\tau) \times (\epsilon, 1-\epsilon)$ for every $\epsilon$, hence throughout $[0,\tau) \times (0,1)$. Applying Itô:

$$\mathrm{d}U_t = \partial_p u(t, P_t)\,\mathrm{d}P_t = \partial_p u(t, P_t)\,\sigma_t(P_t)\,\mathrm{d}W_t^{\mathbb{Q}}.$$

Define the candidate replicating strategy by

$$\theta_s^U = \frac{\phi_s}{\partial_p u(s, P_s)\,\sigma_s(P_s)}, \qquad \theta_s^P = -\theta_s^U\,\partial_p u(s, P_s),$$

which is well-defined on $\{(s,\omega) : P_s(\omega) \in (0,1)\}$, a set of full measure on $[0,\tau] \times \Omega$ since $P_s \in (0,1)$ a.s. for $s < T$. The wealth process $X_t = X_0 + \int_0^t \theta_s^P \,\mathrm{d}P_s + \int_0^t \theta_s^U \,\mathrm{d}U_s$ then satisfies

$$\mathrm{d}X_t = (\theta_s^P + \theta_s^U\,\partial_p u(s, P_s))\sigma_s(P_s)\,\mathrm{d}W_s^{\mathbb{Q}} = \phi_s\,\mathrm{d}W_s^{\mathbb{Q}}$$

by direct substitution. Hence $X_t = \mathbb{E}^{\mathbb{Q}}[H] + \int_0^t \phi_s \,\mathrm{d}W_s^{\mathbb{Q}} = M_t$, and $X_\tau = H$.

The square-integrability of the strategy is the substantive issue, because $\sigma_s(P_s) = P_s(1 - P_s)\beta(s)\gamma(s)^{-1/2}$ vanishes at the boundary $\{P_s \in \{0,1\}\}$ as $s \to T$, and the naive single-asset strategy $\theta^P = \phi_s/\sigma_s(P_s)$ would fail admissibility there. We treat this via a stopping-time approximation.

Define $\tau_\epsilon := \inf\{s \leq \tau : P_s \notin (\epsilon, 1-\epsilon)\} \wedge \tau$ for $\epsilon \in (0,1/2)$, the first exit of $P$ from $(\epsilon, 1-\epsilon)$ within $[0,\tau]$. On $[0,\tau_\epsilon]$, both $P_s$ and $\partial_p u(s, P_s)$ are bounded away from their respective boundaries: $P_s \in [\epsilon, 1-\epsilon]$ by definition, and $\partial_p u$ is bounded above and below on $[0,\tau] \times [\epsilon, 1-\epsilon]$ by continuity of $u$ together with the sign-propagation argument given above (which gives $|\partial_p u| \geq \inf_{p \in [\epsilon, 1-\epsilon]} |h'(p)| > 0$ uniformly in $t$).

The stopped strategy $(\theta_s^P, \theta_s^U)\mathbf{1}_{\{s \leq \tau_\epsilon\}}$ is bounded and predictable, hence admissible, and replicates $M_{\tau_\epsilon}$ on $[0,\tau_\epsilon]$ by direct substitution. Two convergence statements suffice:

*(i) $\tau_\epsilon \to \tau$ a.s. as $\epsilon \downarrow 0$.* For $s < T$, $P_s \in (0,1)$ a.s. (Lemma 3.1 gives $L_s \in \mathbb{R}$ a.s.). Since $\tau < T$ and $P$ has continuous paths on $[0,\tau]$ a.s., $\inf_{s \leq \tau} P_s \wedge \inf_{s \leq \tau}(1 - P_s) > 0$ a.s. on each $\omega$, so for almost every $\omega$ there exists $\epsilon(\omega) > 0$ such that $\tau_\epsilon(\omega) = \tau$ for all $\epsilon < \epsilon(\omega)$.

*(ii) $L^2$-convergence of stopped wealth.* Let $X_t^{(\epsilon)} := X_0 + \int_0^{t\wedge\tau_\epsilon} \phi_s \, \mathrm{d}W_s^{\mathbb{Q}}$ and $X_t = X_0 + \int_0^t \phi_s \, \mathrm{d}W_s^{\mathbb{Q}}$. Then

$$\mathbb{E}[(X_\tau^{(\epsilon)} - X_\tau)^2] = \mathbb{E}\left[\int_{\tau_\epsilon}^{\tau} \phi_s^2 \, \mathrm{d}s\right] \leq \mathbb{E}\left[\int_0^{\tau} \phi_s^2 \, \mathrm{d}s \cdot \mathbf{1}_{\{\tau_\epsilon < \tau\}}\right].$$

By dominated convergence (the integrand is bounded by $\int_0^\tau \phi_s^2 \mathrm{d}s \in L^1$ and $\mathbf{1}_{\{\tau_\epsilon<\tau\}} \to 0$ a.s. by (i)), the right side vanishes as $\epsilon \downarrow 0$. Hence $X_\tau^{(\epsilon)} \to X_\tau = M_\tau = H$ in $L^2(\mathbb{Q})$.

The stopped strategies $(\theta^P \mathbf{1}_{\{s\leq\tau_\epsilon\}}, \theta^U \mathbf{1}_{\{s\leq\tau_\epsilon\}})$ are admissible, replicate $X_\tau^{(\epsilon)}$, and $X_\tau^{(\epsilon)} \to H$ in $L^2$. The limiting strategy obtained by passing to the limit through the Itô integral (using that the integrand $\phi_s$ on $[0,\tau]$ is the same in all $\epsilon$, only the upper limit moves) is the desired admissible replicating strategy, with the boundary set $\{(t,\omega) : t > \tau_\epsilon(\omega) \, \forall\, \epsilon\}$ a $\mathbb{Q}$-null set by (i). Uniqueness up to indistinguishability follows from the uniqueness of $\phi$ in the MRT. □

# B Empirical Methodology

This appendix specifies the methodology for the empirical evaluation of Section 6. The data-collection script and analysis pipeline are released alongside the paper as standalone Python files using only the standard library plus NumPy/SciPy.

## B.1 Data acquisition

The Polymarket Gamma API endpoint `/markets` is queried with parameters `closed=true`, `order=volumeNum`, `ascending=false` to enumerate resolved binary markets ranked by trading volume. For each market that satisfies our minimum-volume filter ($500,000), we extract the YES outcome's CLOB token ID and query `/prices-history?market=<id>&interval=max&fidelity=<f>` on the CLOB API, where the requested fidelity $f$ (in minutes) is selected by cascading through $\{1, 10, 60, 360, 720, 1440\}$ and accepting the finest resolution that returns at least 50 ticks. The CLOB endpoint silently returns an empty history when the requested fidelity is finer than what is retained for a particular market; the cascade extracts data at whatever resolution is available rather than producing false negatives.[1] A topic-based deduplicator (at most six markets per topic, ten markets per resolution week) is applied at the candidate stage to enforce diversity. All API calls are unauthenticated.

## B.2 Effective resolution time

For each market we determine the effective resolution time $T_{\mathrm{eff}}$ as the earliest tick after which the YES price stays within 0.01 of its terminal value. This handles the (empirically common) case of a market that pins effectively several days before the contractual end date.

## B.3 Estimation of $\beta_0$ by realized quadratic variation

Within the data window $[t_0, T_{\mathrm{eff}} - \delta]$ (with $\delta = 5$ minutes), we compute the realized quadratic variation $\widehat{QV} = \sum_i (L_{t_{i+1}} - L_{t_i})^2$ of the logit-transformed price $L_t = \log(P_t/(1-P_t))$ and set

$$\hat{\beta}_0 = \sqrt{\widehat{QV}/(\Lambda_{T_{\mathrm{eff}}-\delta} - \Lambda_{t_0})}, \qquad \Lambda_t := \log \frac{T_{\mathrm{eff}}}{T_{\mathrm{eff}} - t}.$$

The 5-minute buffer prevents the singularity of $\Lambda_t$ at $T_{\mathrm{eff}}$ from polluting the estimator. This is the natural estimator from the bridge SDE: under the calibration (13) with conditional Gaussian law given by Lemma 3.1, $\langle L\rangle_{[t_0,t_1]} = \beta_0^2(\Lambda_{t_1} - \Lambda_{t_0})$ exactly, so $\widehat{QV}/(\Lambda_{t_1} - \Lambda_{t_0})$ is a direct sample from $\beta_0^2$ rather than an estimate of an upper-bound exponent.

[1] See `https://github.com/Polymarket/py-clob-client/issues/216` for documentation of this retention behavior.

### B.4 Estimation of $\hat{\Sigma}_{\text{realized}}$

For forward horizon $\Delta \in \{0.5, 1, 2\}$ days, we form all $(t, t+\Delta)$ pairs with $t+\Delta < T_{\text{eff}}$ where the tick at $t+\Delta$ exists within $\pm 2$ hours of the target time. Pairs are bucketed by time-to-resolution $T_{\text{eff}} - t$ (bins $[0, 0.5), [0.5, 1.5), [1.5, 3), [3, 5), [5, 10), [10, 20)$ days) and by the price anchor $P_t$ (bins $[0.05, 0.30), [0.30, 0.50), [0.50, 0.70), [0.70, 0.95)$). Within each $(T_{\text{eff}} - t, P_t)$ cell we compute, for each strike offset $K - p_{\text{anchor}} \in \{-0.10, -0.05, 0, +0.05, +0.10\}$:

$$\hat{c} = \frac{1}{N} \sum_{i=1}^{N} (P_{t_i+\Delta} - K)^+, \quad \hat{\sigma}_{\text{imp}} = c_{BS}^{-1}(\hat{c}; p_{\text{anchor}}, K, \Delta), \quad \hat{\Sigma}_{\text{realized}} = \hat{\sigma}_{\text{imp}} \sqrt{\Delta},$$

where $c_{BS}^{-1}$ is computed by Brent's method on $[10^{-4}, 200]$ with tolerance $10^{-8}$. Cells with fewer than 10 pairs are dropped. As discussed at the start of Section 6, this object is a Black–Scholes inversion of the realized-payoff sample mean, not of a traded option price, and converges to $\Sigma^*(p, K)$ under the model in the joint limit (28).

### B.5 Cross-market pooling

Per-cell ratios $\hat{\Sigma}_{\text{realized}} / \Sigma^*(p_t, K)$ are pooled across the 50-market panel within each (time-to-resolution bin, strike offset) cell. Within each market, all $(t, t+\Delta)$ pairs with $t + \Delta < T_{\text{eff}}$ are formed (with the temporal tolerance set to $\max(2\,\text{hours}, 1.5 \times \text{median tick spacing})$ so that markets at 720-minute fidelity still yield pairs), and the per-market cell ratio is computed by averaging the realized payoff across the cell's pairs and inverting Black–Scholes against the cell's mean anchor and strike. Pooling across markets is then a simple unweighted mean of the per-market cell ratios, with cross-market standard error reported when at least two markets contribute. Cells with fewer than three pairs in any single market are dropped. Table 2 reports these pooled means, standard errors, and contributing-market counts at $\Delta = 1$ day. Larger $\Delta$ values produce qualitatively similar patterns; the choice $\Delta = 1$ day balances tick coverage against the underlying-payoff variance.

### B.6 Smile-comparison Monte Carlo

For the smile comparison of Table 4, the Jacobi parameter is fit by $\hat{\sigma}_J^2 = \sum_i (\Delta P)_i^2 / \sum_i \overline{P_i(1-P_i)} \Delta t_i$, the realized-variance estimator implied by $\mathrm{d}P = \sigma_J \sqrt{P(1-P)}\mathrm{d}W$.[2] The logit jump-diffusion parameters are fit by a 95th-percentile threshold separating the diffusion increments of $L_t$ from jumps. The fitted Jacobi model is simulated forward from $(t_0, p_0)$ to $T_{\text{eff}}$ using 100,000-path antithetic Monte Carlo with 200 time steps and $[0, 1]$-clipping after each Euler step. The fitted logit JD model is simulated forward in the logit space with 100,000 paths and 200 time steps for the diffusion component plus a single Poisson-thinned jump contribution at the terminal time, valid because the marginal of $L_T$ is unaffected by the jump-time distribution given the total number of jumps in $[t_0, T_{\text{eff}}]$. To enforce the martingale property under our risk-neutral pricing convention, a constant additive shift $c^*$ is applied to the simulated $L_T$ so that $\widehat{\mathbb{E}}[\text{logit}^{-1}(L_T + c^*)] = p_0$ exactly. Without this recentering the logit JD's diffusion drift produces a Jensen bias that mechanically pushes $\widehat{\mathbb{E}}[P_T]$ off $p_0$, contaminating call prices at every strike.

For each model we then compute call prices $\widehat{c}_{\text{Jac}}(K) = \widehat{\mathbb{E}}[(P_T - K)^+]$ and $\widehat{c}_{\text{JD}}(K) = \widehat{\mathbb{E}}[(P_T - K)^+]$ at the strikes shown, and invert the Black–Scholes formula at maturity $T_{\text{eff}} - t_0$ to recover

[2] The fitted $\hat{\sigma}_J \approx 10^{-3}$ values reported in Table 4 are dramatically lower than the synthetic-calibration values of order 1 used in Section 4's exposition. This reflects misspecification: the realized-variance estimator under the $\sqrt{P(1-P)}$ scaling treats sharp news jumps as continuous diffusion, and the boundary-attainment behaviour of Jacobi means most of the realized variance is absorbed by the increasing local volatility as $P_t$ moves away from the mid-point. Either the $\sqrt{P(1-P)}$ specification is wrong for these markets (the data are jumpy, not smoothly diffusive), or the realized-variance moment matching is the wrong calibration target. Both are plausible. The bridge model has no analogous misspecification because $\beta_0$ is fitted directly to the quadratic variation of the logit, where the bridge structure is exactly Gaussian.

the corresponding $\Sigma$. The bridge column reports the closed-form constant $\Sigma^*(p_0, K)$ from (26)–(27), which does not require simulation.

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